From Microscopic Structural Interdependence to Causal Dispersive Geometry

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Abstract

We present an effective, causal completion of General Relativity in which spacetime curvature cannot adjust instantaneously to microscopic change. The driving ingredient is a single scalar "structural activity" observable built from quantum fluctuation rate, entropy, and correlations (the Quantum Structural Triad). When this structural activity persists for a finite time—as measured by its autocorrelation—standard linear-response principles imply that the gravitational coupling must become mildly frequency-dependent and exhibit a universal near-static delay. In this sense, dispersive gravity is not added by hand: it is forced by finite microscopic persistence together with causality and passivity. For practical use we also discuss a minimal one-timescale relaxation closure that yields a simple first-order time-lag law for curvature along a local coarse-graining flow, and we outline a covariant Closed-Time-Path construction that maintains conservation in the linear-response setting. The resulting relaxation time sets characteristic length and inertial scales, recovering the Einstein limit for slowly varying sources while predicting causal attenuation at higher frequencies. This manuscript provides the Green–Kubo/Kramers–Kronig necessity foundation underlying the phenomenological geometric-response framework of Ref. [1].

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1 Introduction

Standard General Relativity (GR) operates on the assumption that spacetime geometry responds *instantaneously* to changes in matter distributions [2, 3]. This implies that the "structural rearrangement" of the manifold occurs at infinite speed, creating a conceptual tension with microscopic causality. In the strict Einstein (DC) limit, the coupling G is nondispersive and has no dependence on microscopic time scales. Yet realistic quantum matter never evolves without characteristic speeds and relaxation times: locality enforces Lieb–Robinson lightcones, interacting media relax on finite microscopic windows, and linear-response theory generically yields dispersive susceptibilities [4]. If curvature is sourced—directly or indirectly—by such finite-rate microscopic activity, then its response cannot be arbitrarily fast. Causal effective descriptions of gravitation permit retarded, analytic, passive kernels; retarded Green functions are standard in covariant linear response consistent with Ref. [2]. This raises the central question of this work:

How does the persistence of microscopic structural fluctuations impose necessary constraints on the macroscopic geometric response?

A structural answer begins with the observation that geometry cannot reorganize faster than the microscopic dynamics can propagate, relax, and dissipate their fluctuations. On any band where microscopic dynamics exhibits finite correlation, the response must contain a finite causal delay. The key is to connect this delay to a provable microscopic quantity. To that end, we follow the necessity chain:

$$\Pi(t) \implies C_\Pi(t) \implies \tau_G = \int_0^\infty C_\Pi(t)\,dt \implies \chi(\omega) \approx 1 - i\omega\tau_G \implies G_{\text{eff}}(\Box) = G_0\,\chi(\Box).$$

Scope note. Here $\chi(\Box)$ denotes a *retarded* covariant kernel/operator encoding causal memory; the KK argument fixes only its universal *low-frequency* behaviour (slope), while specific covariant realizations (e.g. slow-band representatives along a local coarse–graining flow) are modelling choices beyond the necessity result.
where each arrow is governed by a standard principle of linear response: finite autocorrelation $\Rightarrow$ finite Green–Kubo (GK) time; causality and passivity $\Rightarrow$ Kramers–Kronig (KK) analyticity; KK analyticity $\Rightarrow$ universal low-frequency slope. This sequence establishes that the dispersive character of gravity is not an ad hoc modification, but an unavoidable consequence of the source's temporal structure.

Structural provenance from previous work. The structural observable $\Pi(t)$ originates in the companion manuscript by O. Tataru, *A Quantum Structural Triad: Fluctuations, Entropy, and Correlations as Interdependent Primitives* (Int. J. Quantum Foundations **11** (4), 800–852, 2025) [5]. For any subsystem A in a finite-range, locally coupled system, that work introduced:

$$\Pi_A(t) = \sqrt{F_Q(\rho_A(t); H_A)}\, S_A(t)\, I(A{:}\bar{A}, t), \tag{1}$$

a scalar that merges fluctuation strength (Quantum Fisher Information, QFI), entropic multiplicity, and cross-cut correlation into a single measure of *structural persistence*. Under hypotheses H1–H3 (local finite-range interactions, genuine cross-cut coupling, and a locally weakly stationary and mixing epoch), the three legs cannot fluctuate independently: a persistent deficit in one requires compensatory activity in the others, while simultaneous collapse marks a regime of structural

inertness. Ref. [5] proved that $\Pi(t)$ cannot vanish on sets of nonzero measure, a result guaranteed by the Paley–Zygmund inequality which applies rigorously to processes with finite variance and positive support under windowed averaging. Although defined in subsystem language, the triad is systemic: the partition acts purely as a diagnostic of global structural integrity.

A second companion manuscript by O. Tataru, *Geometric Response and the Frequency-Dependent Gravitational Coupling* (Int. J. Quantum Foundations **11** (4), 853–905, 2025) [1], developed the phenomenology and calibration of slow-band frequency-dependent coupling and adopted the scalar relaxation law

$$\tau_G \dot{R} + R = G_0 R_{\text{info}} \tag{2}$$

as an *effective ansatz*. That work emphasized operational tests, closures (Debye and multipole), and numerical controls.

The present manuscript is a theoretical sequel to Ref. [1]: it supplies the missing *necessity* result that constrains any admissible causal geometric-response kernel. Under causality, passivity, and DC regression/Fluctuation-Dissipation Theorem (FDT) normalization, the Green–Kubo correlation time of Π *fixes* the universal small-frequency slope of the geometric susceptibility via Kramers–Kronig analyticity. In particular, the operator form $G_{\text{eff}}(\Box) = G_0\,\chi(\Box)$ and its low-frequency expansion arise from the microscopic structural dynamics of Π, rather than from assuming a specific closure such as a single-pole (Debye) kernel. This places the phenomenology of Ref. [1] on a rigorous structural/linear-response foundation while leaving high-frequency completion as a modelling choice.

Physical meaning of "information" and R_{info}. Throughout this work, we use the word "information" in a strictly physical, non-semantic sense: it quantifies structural organization and dynamical complexity, not a symbolic or linguistic content. The informational curvature source R_{info} is therefore *not* a Shannon bit count; it is a scalar built from the triad observable and has the dimensions of a curvature or acceleration scale, $[R_{\text{info}}] = T^{-2}$.

Following the companion triad paper [5], we view the legs of the triad as a **Rate/Load decomposition**:

- **The Rate:** The quantum Fisher information $\sqrt{F_Q}$ sets the *driving frequency* of structural evolution (the reactive term).
- **The Load:** The entropy S and mutual information I quantify the *structural viscosity* (the dissipative term)—the density of degrees of freedom that must be reorganized.

This decomposition aligns with linear response in viscoelastic media, distinguishing the inertial cost of change from the entropic resistance to flow. Concretely, in the slow-band constitutive law $\tau_G \dot{R} + R = G_0 R_{\text{info}}$, we take R_{info} to be calibrated from this structural activity. Thus, R_{info} plays the role of a *structural power density* required to maintain the quantum state. When the microscopic driving rate set by $\sqrt{F_Q}$ approaches or exceeds the geometric relaxation rate τ_G^{-1}, the resulting relaxational (Maxwell–Cattaneo/Debye) dynamics implies that high-frequency geometric modes are causally attenuated rather than able to follow the informational driving instantaneously.

Microscopic delay and the Green–Kubo time. On any locally weakly stationary and mixing window (H3), fluctuations of $\Pi(t)$ cannot decorrelate instantly; relaxation occurs only after information has propagated across the local Lieb–Robinson lightcone. This enforces a finite microscopic Green–Kubo time:

$$\tau_G = \int_0^\infty C_\Pi(t)\,dt, \tag{3}$$

defined wholly in terms of the structural observable and without any geometric assumptions. This finite integration time captures the equilibration scale of the underlying quantum microstructure.

Causal inheritance of finite responsiveness. If curvature is sourced—even indirectly—through an observable encoding this microscopic structural activity, then geometry must inherit the same finite delay. Any causal relation between a structural driver and a geometric response is encoded in a retarded kernel; such kernels are analytic and passive, and their susceptibilities obey Kramers–Kronig relations. The remaining question is whether the kernel's form can be *derived* from the structural properties of $\Pi(t)$.

Main result. We now summarize the main necessity statement. Under causality, passivity, and regression/FDT normalization, KK analyticity forces the universal small-frequency form:

$$\chi(\omega) = 1 - i\,\omega\,\tau_G + O(\omega^2) \quad (|\omega|\,\tau_G \ll 1), \qquad G_{\text{eff}}(\omega) = G_0\,\chi(\omega). \tag{4}$$

Thus, the relaxation time extracted from Π directly *controls* the low-frequency slope of the effective coupling. No constitutive assumption is needed: the form of G_{eff} follows from GK scaling plus causal linear response. When a minimal closure is convenient, the one-pole expression $(1+i\omega\tau_G)^{-1}$ serves as a *slope-saturating* closure, consistent with Mori–Zwanzig projection arguments for single-mode truncation, rather than as a fundamental model.

Crucially, this derivation relies solely on the premises H1–H3 (defined in Sec. 1); no additional microscopic or thermodynamic hypotheses are introduced beyond the structural definition of the source.

Theorem 1 (Low-frequency susceptibility (necessity))**.** ***Hypotheses:*** *On an LWSS/mixing window (H3) with locality and genuine coupling (H1–H2), suppose:*

(i) $C_\Pi \in L^1([0,\infty))$ *so* $\tau_G := \int_0^\infty C_\Pi(t)\,dt$ *exists;*

(ii) the retarded kernel $R(t)$ *is causal,* $R \in L^1$*, and* $tR \in L^1$*;*

(iii) passivity $\Im\chi(\omega) \le 0$ *for* $\omega > 0$*;*

(iv) regression/FDT calibration $\chi(0) = 1$*.*

Conclusion: *As* $\omega \to 0$*,*

$$\chi(\omega) = 1 - i\,\omega\,\tau_G + O(\omega^2), \qquad G_{\text{eff}}(\omega) = G_0\,\chi(\omega).$$

Sketch. Causality $\Rightarrow$ upper-half-plane analyticity; $tR \in L^1$ enables KK differentiation at $\omega = 0$; FDT fixes $\chi(0) = 1$; KK fixes the slope moment $\partial_\omega \Im\chi|_0$ and regression/FDT identifies it with $-\tau_G$. See Appendix D (derivations: GK→KK, FDT, passivity, and short-time bounds).[1]

[1]Formal coverage and scope of the GK→KK chain and Debye sufficiency are summarized in Appendix E (Lean project modules).

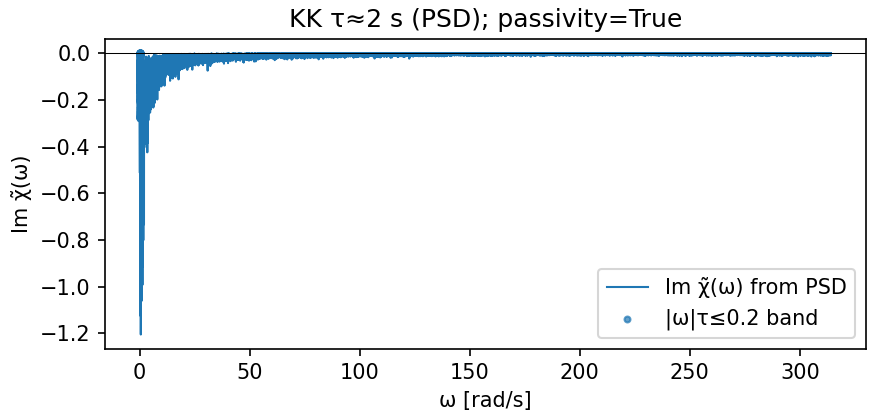


Figure 1: OU baseline (slow band): near-DC slope estimate $\tau_{\mathrm{KK}} = -\Im\chi(\omega)/\omega$ matches the GK value τ_G within tolerance; passivity holds ($\Im\chi(\omega) \leq 0$ for $\omega > 0$). Reproducibility parameters and extended diagnostics appear in Appendix B/E.

Kinematic scales. A finite τ_G defines a characteristic frequency $\omega_G = \tau_G^{-1}$, separating quasi-Einsteinian ($\omega\tau_G \ll 1$) from dispersive response. In relativistic units, this sets an inertial scale:

$$m_G = \frac{\hbar}{c^2\tau_G}, \tag{5}$$

together with $\ell_G = c\,\tau_G$ and $\lambda_G = 2\pi c\,\tau_G$, which delimit the band on which curvature can adiabatically track its structural drivers. Equivalently, a causal covariantization of the Debye relaxation closure is expressed along a local timelike coarse–graining flow u^μ as

$$G_{\mathrm{eff}} = G_0\big(1 - \tau_G\, u^\mu\nabla_\mu\big)^{-1} = G_0\Big[\,1 + \tau_G\, u^\mu\nabla_\mu + O((\tau_G\nabla)^2)\Big], \tag{6}$$

whose Fourier symbol in the local rest frame is $G_{\mathrm{eff}}(\omega) = G_0/(1 + i\omega\tau_G)$ and thus matches the universal KK slope. The condition $|\omega|\tau_G \ll 1$ then directly characterizes the adiabatic band.

Interpretation. Geometry behaves as a causal, finite-rate medium whose low-frequency response is governed by the persistence of microscopic structural activity. The Einstein regime emerges on slow bands ($G_{\mathrm{eff}} \to G_0$), while dispersive attenuation appears when $\omega\tau_G \gtrsim 1$. Crucially, this finite-rate response arises from the medium's inertia, not from the addition of new propagating degrees of freedom (e.g., massive graviton fields) within the slow band description. The derivation provided here is therefore not a mechanical modification of GR but a structural consequence of finite-rate microscopic dynamics.

Background and placement

This work operates within a standard covariant linear-response/EFT setting [2] with dispersive susceptibilities [4] and CTP (Schwinger–Keldysh) causal actions used to ensure exact conservation. It draws on geometric thermodynamics ideas insofar as static normalization identifies the adiabatic stiffness ($\chi(0) = 1$, fixing G_0), but it distinguishes this clearly from the dynamic susceptibility ($\chi(\omega)$, governing dispersion) which does not require positing new thermodynamic degrees of freedom.

Microscopically, the triad Π packages QFI-based speed, entropy, and mutual information into a single structural diagnostic; locality is controlled by Lieb–Robinson lightcones; LWSS/mixing

windows provide the operational regime where GK integrals and KK expansions are well-behaved. The novelty lies in the necessity chain: finite microscopic persistence $\Rightarrow$ finite GK time $\Rightarrow$ KK-constrained slope, so that $G_{\text{eff}}(\omega) = G_0\,\chi(\omega)$ inherits its small-ω behaviour from τ_G. We defer optional multipole generalizations to future work and keep conceptual claims conservative to comply with APS/COPE expectations.

Organization. The manuscript is structured as a microscopic $\to$ geometric $\to$ testing chain rather than as independent topics. First, Structural Triad (Preliminaries and Definitions) recalls the definition of the structural triad Π and the standing hypotheses (H1–H3). The microscopic development in Microscopic Dynamics of the Structural Triad (together with its Green–Kubo continuation) derives the explicit time evolution of $\Pi(t)$, its autocorrelation $C_\Pi(t)$, and the Green–Kubo relaxation time τ_G, supported by OU and Lindblad examples. Geometric Response and Frequency–Dependent Coupling then lifts this scalar relaxation time into a covariant geometric response: it formulates the informational curvature source R_{info}, derives the KK-constrained low-frequency behaviour of $G_{\text{eff}}(\omega) = G_0\chi(\omega)$ (and representative covariant realizations), and constructs the CTP action ensuring exact conservation (Sec. 15.8.1). Finally, Discussion and Scope discusses the domain of validity, provides a concrete GK $\to$ KK testing protocol, and summarizes how the microscopic structural conditions for quantum viability determine the macroscopic causal structure of geometric response. Appendices collect the microscopic derivations and technical sketches for τ_G and m_G, numerical protocol and diagnostics (with Mori–Zwanzig view and checks), and formal verification.

Assumptions and scope (main text). H1–H3 (local finite-range locality; genuine cross-cut coupling; windowed LWSS+mixing); causality (retarded support) and passivity ($\Im\chi(\omega) \le 0$, $\omega > 0$); DC regression/FDT normalization ($\chi(0) = 1$); integrability conditions where used ($C_\Pi \in L^1$, $R, tR \in L^1$ for KK slope); slow-band regime $|\omega|\,\tau_G \ll 1$; statements are *necessary* (not sufficient). Debye is optional and slope-saturating.

In this framework, we do not assume any *specific constitutive kernel* beyond retarded linear response, passivity, and DC normalization; given a triad-based structural driver sourcing curvature in linear response, the GK$\to$KK chain then fixes the universal near-DC slope without requiring a Debye (single-pole) closure. If one adopts the thermodynamic interpretation of gravitational dynamics—where the local Clausius relation $\delta Q = T, \delta S$ on causal horizons underlies the Einstein equations [6]—then the relevant source must supply both an entropy density and a physically meaningful fluctuation rate. The structural triad $\Pi(t)$ provides precisely these ingredients: it combines driving frequency and organizational load into a single scalar with the dimensions of structural power. Once $\Pi(t)$ exhibits a finite Green–Kubo correlation time, causality and Kramers–Kronig analyticity compel a finite geometric susceptibility, with τ_G fixing the low-frequency slope of $G_{\text{eff}}(\omega)$. Thus, the triad–geometry coupling arises *necessarily* from the persistence of microscopic structural fluctuations under H1–H3 and linear-response conditions; no independent interaction postulate is required.

Part I
Structural Triad (Preliminaries and Definitions)

Provenance and scope. This Part summarizes the structural input required for the Green–Kubo and Kramers–Kronig development of the geometry–relaxation time τ_G and the effective susceptibil-

ity $G_{\text{eff}}(\omega)$. Only the definitions and standing hypotheses introduced in the companion manuscript *A Quantum Structural Triad: Fluctuations, Entropy, and Correlations as Interdependent Primitives* [5] are recalled here. Full proofs of the triad's systemic interdependence, non-vanishing, and threshold bounds are contained in [5] and are not reproduced.

2 Structural Input: The Quantum Triad

2.1 Definition of the structural observable

For a subsystem A with reduced state $\rho_A(t)$ and local generator H_A, the *Structural Triad* is the scalar observable

$$\Pi_A(t) = \sqrt{F_Q(\rho_A(t); H_A)}\; S_A(t)\; I(A{:}\bar{A}, t). \tag{7}$$

The three factors encode complementary structural aspects:

- $\sqrt{F_Q(\rho_A; H_A)}$: quantum Fisher information (time encoding $H_A/\hbar$) with units [s^{-1}];
- S_A: von Neumann entropy (nats);
- $I(A{:}\bar{A})$: mutual information between A and its complement (nats).

Thus $[\Pi_A] = \mathrm{s}^{-1}$ and the cut–average $\Pi(t) = \frac{1}{|E|}\sum_{A\in E}\Pi_A(t)$ is a fluctuating indicator of microscopic structural activity.

2.2 Standing hypotheses (H1–H3)

We adopt the operational assumptions established in [5]:

(H1) Locality. Interactions have finite range and bounded degree, so Lieb–Robinson lightcones constrain information propagation.

(H2) Genuine cross-cut coupling. The bipartition $(A|\bar{A})$ is crossed by non-trivial interaction terms; $I(A{:}\bar{A})$ does not vanish identically.

(H3) Windowed LWSS + mixing. On a time window W, the process is locally weak-sense stationary (finite first/second moments, stable mean and variance) and mixing, so the normalized autocorrelation $C_\Pi(t)$ is absolutely integrable.

These conditions ensure that $\Pi(t)$ is continuous on W, possesses a well-defined power spectral density, and admits a finite Green–Kubo integral.

2.3 Minimal dynamic assumptions (R1–R4)

For use in later sections (operator form and KK conditions) we recall the microscopic regularity assumptions:

(R1) Locality. Same as H1; Lieb–Robinson bounds ensure quasi-locality of evolved observables.

(R2) Full-rank regularity. On W, $\rho_A(t)$ is full rank and piecewise C^1; hence S_A, $I(A{:}\bar{A})$, and $\sqrt{F_Q}$ are continuous.

(R3) Clustering/mixing. Exponential clustering (Gibbs/KMS) or primitive GKLS mixing, ensuring $C_\Pi \in L^1(\mathbb{R}_+)$.

(R4) Bounded quasi-locality. $\Pi_A(t)$ remains bounded on compact epochs and quasi-local in the Lieb–Robinson sense.

Under R1–R4, $\Pi(t)$ forms a well-posed stochastic process with a finite correlation time.

2.4 The necessity chain (structural → geometric)

Under the above hypotheses, the structural observable $\Pi(t)$ determines the low-frequency behaviour of the geometric susceptibility through the standard linear-response chain:

$$\Pi(t) \implies C_\Pi(t) \implies \tau_G = \int_0^\infty C_\Pi(t)\,dt \implies \chi(\omega) = 1 - i\omega\tau_G + O(\omega^2) \implies G_{\text{eff}}(\omega) = G_0\,\chi(\omega).$$

Each arrow corresponds to a standard principle of non-equilibrium statistical mechanics: Green–Kubo (GK) for the transport coefficient τ_G, and Kramers–Kronig (KK) for the causal dispersion of $\chi(\omega)$.

2.5 Green–Kubo relaxation time

The microscopic persistence of structural fluctuations is encoded in the normalized autocorrelation

$$C_\Pi(t) = \frac{\langle \delta\Pi(0)\delta\Pi(t)\rangle}{\langle(\delta\Pi(0))^2\rangle}, \qquad C_\Pi(0) = 1. \tag{8}$$

Under H3/R3 (LWSS + mixing), $C_\Pi \in L^1(\mathbb{R}_+)$ and the Green–Kubo geometry–relaxation time

$$\tau_G := \int_0^\infty C_\Pi(t)\,dt \tag{9}$$

exists and is finite. It quantifies the persistence of microscopic structural fluctuations and sets the causal time scale below which geometry cannot respond.

2.6 Kramers–Kronig necessity at low frequency

Let $R(t)$ be the retarded kernel relating the informational driver to the geometric curvature, and let $\chi(\omega)$ be the associated susceptibility. Throughout we use the one-sided Fourier–Laplace convention $\chi(\omega) = \int_0^\infty R(t)\,e^{-i\omega t}\,dt$, so that a Debye kernel yields $\chi(\omega) = (1 + i\omega\tau_G)^{-1}$ and passivity reads $\Im\chi(\omega) \le 0$ for $\omega > 0$. Under: (i) causality ($R(t < 0) = 0$), (ii) passivity ($\Im\chi(\omega) \le 0$ for $\omega > 0$), (iii) absolute integrability ($R, tR \in L^1$), and (iv) regression/FDT normalization ($\chi(0) = 1$), KK analyticity yields the universal small-frequency form:

$$\chi(\omega) = 1 - i\,\omega\,\tau_G + O(\omega^2), \tag{10}$$

so that

$$G_{\text{eff}}(\omega) = G_0\,\chi(\omega) = G_0\big(1 - i\omega\tau_G + O(\omega^2)\big). \tag{11}$$

These statements are *necessary* consequences of GK+KK under the specified hypotheses; no constitutive assumptions are imposed.

Regression/FDT bridge (identifying the slope with τ_G). The KK slope is controlled by the first moment of the retarded kernel: if $tR \in L^1(\mathbb{R}_+)$ then $\partial_\omega \Im\chi(\omega)\big|_{\omega=0} = -\int_0^\infty t\,R(t)\,dt$. To identify this moment with the Green–Kubo time $\tau_G = \int_0^\infty C_\Pi(t)\,dt$ one uses the regression/FDT relation between response and correlation. In the classical regression form (for a normalized, stationary observable) one has $R(t) = -\partial_t C_\Pi(t)$ (up to standard contact/normalization terms) with $C_\Pi(0) = 1$ and $C_\Pi(t) \to 0$, so integration by parts gives $\int_0^\infty t\,R(t)\,dt = \int_0^\infty C_\Pi(t)\,dt = \tau_G$.

2.7 Validity regime and interpretation

The necessity statements apply on slow bands satisfying $|\omega|\tau_G \ll 1$ and on backgrounds whose curvature radii exceed the screening length $\ell_G = c\,\tau_G$. On such windows, a causal covariantization of the *Debye* relaxation law is naturally written along a local timelike coarse–graining flow u^μ as a retarded operator $(1 - \tau_G\, u^\mu \nabla_\mu)^{-1}$, whose Fourier symbol in the local rest frame is $(1 + i\omega\tau_G)^{-1}$ and hence reproduces the universal KK slope in (10). For bookkeeping on very slow modes one may expand this retarded operator as a gradient (Neumann) series,

$$G_{\text{eff}} = G_0\big(1 - \tau_G\, u^\mu \nabla_\mu\big)^{-1} = G_0\big[1 + \tau_G\, u^\mu \nabla_\mu + O((\tau_G \nabla)^2)\big].$$

The quantity τ_G is therefore the microscopic origin of the macroscopic phase lag and dispersive attenuation of geometric response.

Assumptions and scope (Part I).

- **Microscopic:** H1–H3 (finite-range locality, genuine cross-cut coupling, LWSS+mixing).
- **Regularity:** R1–R4 (continuity, full-rank regularity, clustering/mixing, quasi-locality).
- **Response:** Causality, passivity, and regression/FDT normalization ($\chi(0) = 1$).
- **Integrability:** $C_\Pi \in L^1$ (GK), $R, tR \in L^1$ (KK slope).
- **Regime:** Slow-band $|\omega|\,\tau_G \ll 1$; statements are *necessary*. Debye appears only as a sufficient slope-saturating closure.

Part II
Microscopic Dynamics of the Structural Triad

Scope. This Part derives the explicit time dependence of the structural observable $\Pi_A(t) = \sqrt{F_Q(\rho_A(t); H_A)}\, S_A(t)\, I(A{:}\bar{A}, t)$ from the microscopic evolution of the reduced state $\rho_A(t)$. Unlike Part I—which only fixed definitions, assumptions (H1–H3), and normalizations—this Part provides the dynamical input required for the Green–Kubo construction of the geometry–relaxation time τ_G and the subsequent Kramers–Kronig analysis. No constitutive assumptions are introduced here; all results follow directly from the quantum equations of motion.

3 Microscopic Equations of Motion

We consider a bipartition $(A|\bar{A})$ of a finite quantum system with Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_{\bar{A}}$. The global evolution is generated either by a Hamiltonian or by a local GKLS generator. Both cases yield a well-defined reduced evolution $\rho_A(t)$.

3.1 Closed dynamics

If the full system evolves unitarily,

$$\dot{\rho}(t) = -\frac{i}{\hbar}[H, \rho(t)], \qquad \rho(t) = e^{-iHt/\hbar}\,\rho(0)\,e^{iHt/\hbar}, \tag{12}$$

then the reduced state evolves as

$$\rho_A(t) = \mathrm{Tr}_{\bar{A}}\Big[e^{-iHt/\hbar}\,\rho(0)\,e^{iHt/\hbar}\Big], \tag{13}$$

and remains differentiable provided $\rho(0)$ is full rank on A.

3.2 Open dynamics

Under a local GKLS equation

$$\dot{\rho}(t) = -\frac{i}{\hbar}[H,\rho] + \sum_j \Big(L_j \rho L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j, \rho\} \Big)\,, \tag{14}$$

one has the reduced evolution

$$\rho_A(t) = \mathrm{Tr}_{\bar{A}}\, \rho(t), \qquad \dot{\rho}_A(t) = \mathrm{Tr}_{\bar{A}}\, \dot{\rho}(t), \tag{15}$$

preserving complete positivity and trace. All subsequent expressions remain valid in either case.

4 Time Dependence of the Triad Components

We collect the evolution equations for the three structural components.

4.1 Entropy

The von Neumann entropy of the reduced state is

$$S_A(t) = -\,\mathrm{Tr}[\rho_A(t)\log\rho_A(t)]\,,$$

with derivative

$$\dot{S}_A(t) = -\,\mathrm{Tr}\big[\dot{\rho}_A(t)\big(\log\rho_A(t) + \mathbb{1}\big)\big]\,. \tag{16}$$

This expression holds for full-rank $\rho_A(t)$ on any LWSS window.

4.2 Mutual information

The mutual information is

$$I(A{:}\bar{A},t) = S(\rho_A) + S(\rho_{\bar{A}}) - S(\rho),$$

so

$$\dot{I}(A{:}\bar{A},t) = \dot{S}(\rho_A) + \dot{S}(\rho_{\bar{A}}) - \dot{S}(\rho). \tag{17}$$

This term captures the flow of correlations across the cut.

4.3 Quantum Fisher information

For a full-rank spectral decomposition $\rho_A = \sum_i p_i |i\rangle\langle i|$, the quantum Fisher information is

$$F_Q(\rho_A; H_A) = 2\sum_{i,j} \frac{(p_i - p_j)^2}{p_i + p_j} \big|\langle i|H_A|j\rangle\big|^2. \tag{18}$$

Its derivative follows from $\dot{p}_i$ and the first-order perturbation of the eigenvectors:

$$\dot{F}_Q = 4\,\Re\!\sum_{i,j} \frac{(p_i - p_j)(\dot{p}_i - \dot{p}_j)}{p_i + p_j} \big|\langle i|H_A|j\rangle\big|^2 + (\text{eigenvector terms}). \tag{19}$$

The eigenvector terms are controlled using Kato perturbation theory.

5 Derivative of the Triad Observable

Combining Eqs. (16), (17), and (19), the time derivative of $\Pi_A(t) = \sqrt{F_Q}\, S\, I$ is

$$\dot{\Pi}_A(t) = \frac{\dot{F}_Q}{2\sqrt{F_Q}}\, S\, I + \sqrt{F_Q}\, \dot{S}\, I + \sqrt{F_Q}\, S\, \dot{I}. \tag{20}$$

All three contributions are computable once $\rho_A(t)$ is known. The sign and magnitude of $\dot{\Pi}_A$ reflect competing tendencies: growth of local uncertainty, redistribution of correlations, and changes in the state's intrinsic speed.

Non-negativity and temporal support. Under the standing hypotheses (H1–H3), $\Pi_A(t)$ is a nonnegative, finite-variance process, continuous on LWSS windows, ensuring the existence of its autocorrelation function.

6 Autocorrelation and Structural Persistence

Define coarse averages over a cut-set E and centered fluctuations

$$\Pi(t) = \frac{1}{|E|} \sum_{A \in E} \Pi_A(t), \qquad \delta\Pi(t) = \Pi(t) - \langle \Pi \rangle.$$

The normalized autocorrelation function (ACF) is

$$C_\Pi(t) = \frac{\langle \delta\Pi(0)\, \delta\Pi(t) \rangle}{\langle (\delta\Pi(0))^2 \rangle}, \qquad C_\Pi(0) = 1. \tag{21}$$

LWSS and mixing (H3) imply:

$$C_\Pi \in L^1(\mathbb{R}_+) \quad \Longrightarrow \quad \tau_G := \int_0^\infty C_\Pi(t)\, dt < \infty.$$

Thus the *persistence time* of structural fluctuations is determined dynamically by the decay rate of $C_\Pi(t)$.

7 Model Example and Analytic Surrogate

For numerical benchmarking, $\Pi(t)$ can be approximated by an Ornstein–Uhlenbeck (OU) surrogate consistent with (H1–H3):

$$d\Pi = -\frac{\Pi}{\tau_0}\, dt + \sigma\, dW_t, \qquad \langle \Pi(t)\Pi(t+\tau) \rangle = \frac{\sigma^2 \tau_0}{2}\, e^{-|\tau|/\tau_0}. \tag{22}$$

In this surrogate $\tau_G = \tau_0$, and the retarded response kernel realizes the Debye form

$$G_{\text{eff}}(\omega) = \frac{G_0}{1 + i\omega\tau_G}.$$

This provides a controlled analytic benchmark for the GK–KK diagnostics.

8 Interpretation

The microscopic evolution of the triplet $\left(\sqrt{F_Q}, S, I\right)$ demonstrates that:

1. $\Pi(t)$ is a dynamically generated observable, not a static invariant.
2. Its fluctuations encode the persistence of microscopic structural activity.
3. The decay of its autocorrelation directly defines the Green–Kubo relaxation time τ_G.

Thus Part II supplies the dynamical bridge from local quantum evolution to the transport-type quantity τ_G, preparing the ground for the causal response and geometric dispersive structure developed in the following sections.

9 Operator form, regularity, and quasi–locality

On full–rank epochs, $t \mapsto S_A(t)$ and $t \mapsto I(A{:}\bar{A}, t)$ are continuous; the SLD metric continuity implies $t \mapsto \sqrt{F_Q(\rho_A(t); H_A)}$ is continuous, hence $t \mapsto \Pi_A(t)$ is continuous (indeed C^1 on C^1 trajectories). Lieb–Robinson bounds yield quasi–locality: first variations of Π_A outside the local Lieb–Robinson lightcone are exponentially small.

Theorem 2 (Regularity and quasi–locality)**.** *Under locality and full–rank regularity, $\Pi_A(t)$ is continuous (and C^1 on C^1 trajectories) and quasi–local in the Lieb–Robinson sense.*

10 Autocorrelation, Green–Kubo time, and susceptibility

Define fluctuations $\delta\Pi = \Pi - \langle \Pi \rangle$ and

$$C_\Pi(t) = \frac{\langle \delta\Pi(t)\, \delta\Pi(0) \rangle}{\langle (\delta\Pi(0))^2 \rangle}, \qquad C_\Pi(0) = 1, \qquad \tau_G = \int_0^\infty C_\Pi(t)\, dt.$$

Under windowed LWSS/mixing (thermal clustering or primitive GKLS), $C_\Pi \in L^1([0,\infty))$ so τ_G is finite. The retarded susceptibility

$$\chi_\Pi(\omega) = \int_0^\infty R_\Pi(t)\, e^{-i\omega t}\, dt$$

is analytic in the upper half–plane and obeys Kramers–Kronig. Here $R_\Pi(t)$ is the retarded response kernel of the triad observable; under regression/FDT one may identify $R_\Pi(t) = -\partial_t C_\Pi(t)$ up to standard contact/normalization terms (see Sec. 2.6). With DC normalization,

$$\chi_\Pi(\omega) = 1 - i\,\omega\,\tau_G + O(\omega^2), \qquad -\partial_\omega \Im \chi_\Pi(\omega)\big|_{\omega=0} = \tau_G.$$

For a unified dispersion statement and its implication for $G_{\text{eff}}(\omega) = G_0\, \chi(\omega)$, see Sec. 2.6, Proposition 1.

Theorem 3 (GK existence)**.** *Under locality and mixing/clustering on full–rank epochs, $C_\Pi \in L^1([0,\infty))$ and τ_G is finite.*

Theorem 4 (KK dispersion at small frequency)**.** *Under causality/passivity and DC normalization, $\chi_\Pi(\omega) = 1 - i\omega\tau_G + O(\omega^2)$ and the KK relations hold between real and imaginary parts.*

11 Conjugate perturbation (triad–leg source)

To operationalize linear response, perturb the Hamiltonian on A by a calibrated source $h(t)$:

$$H \longrightarrow H - h(t)\, X_A, \qquad X_A := \frac{H_A}{\|H_A\|_{\mathrm{op}}}, \qquad \dot{\rho} = -\frac{i}{\hbar}[H - h(t)X_A, \rho].$$

The susceptibility of the triad is

$$\chi_\Pi(t) = \frac{\delta\langle \Pi_A(t)\rangle}{\delta h(0)}\bigg|_{h=0},$$

measuring the response of Π_A to the local generator pulse. Correlation–leg and entropic–leg sources are also admissible in principle (modulating cross–cut couplings or entropy production via GKLS deformations). *Physics note.* Here X_A is a unit-normalized "energy leg" on A, and $h(t)$ is a small, localized perturbation. This is how we physically probe how strongly the structural activity Π reacts to a controlled change in the local dynamics.

12 Triad susceptibility and effective modulus

At the level of the microscopic driver, the response of the structural triad to a small source $h(t)$ is encoded in the *triad susceptibility*

$$\chi_\Pi(\omega) := \int_0^\infty R_\Pi(t)\, e^{-i\omega t}\, dt,$$

which characterizes how the expectation value of $\Pi_A(t)$ changes per unit variation of its conjugate perturbation $h(t)$. In the geometric sector we introduce the *effective modulus*

$$G_{\mathrm{eff}}(\omega) := G_0\, \chi(\omega),$$

where $\chi(\omega)$ is the macroscopic geometric susceptibility constrained by the GK→KK analysis. In this nomenclature:

- $\chi_\Pi(\omega)$ is the susceptibility of the microscopic triad field (strain-like response per unit generalized stress h);
- $\chi(\omega)$ is the scalar susceptibility relating the informational curvature R_{info} to the geometric curvature $R_{\mu\nu}$;
- $G_{\mathrm{eff}}(\omega) = G_0\, \chi(\omega)$ plays the role of a (frequency-dependent) geometric stiffness or modulus.

At low frequency, the geometric susceptibility obeys

$$\chi(\omega) = 1 - i\,\omega\,\tau_G + O(\omega^2),$$

so that $G_{\mathrm{eff}}(\omega)$ inherits the universal KK slope $-\partial_\omega \Im\chi|_{\omega=0} = \tau_G$ while remaining passive and causal. *Notation note.* We reserve the term "modulus" for G_{eff}; χ_Π and χ are always referred to as susceptibilities. For the unified causality/KK statements and DC normalization, see Sec. 2.6; for the optional single-pole closure used in some plots, see Sec. 15.2.

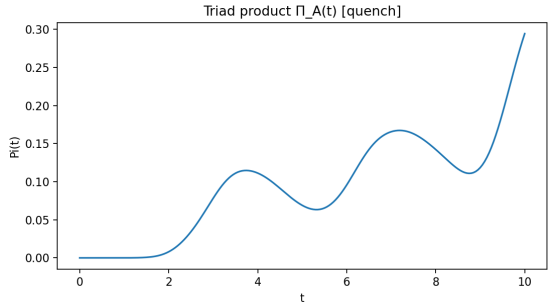
13 Numerical example: XXZ quench and thermal control

We illustrate with TEBD and ED runs (time encoding $H/\hbar$; one–sided PSD).

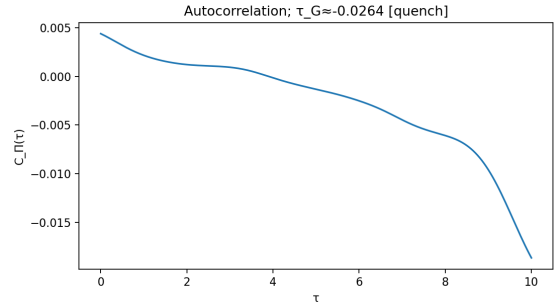
Numerical conventions and caveats.

- QFI leg (pure states): for pure global dynamics we use the standard approximation $\sqrt{F_Q} \approx 2\,\Delta H_A$ for the reduced state, providing a conservative upper estimate of the speed leg.
- Quench vs KMS: quench examples are non-equilibrium and not KMS/Gibbs tests; thermal (KMS) snapshots serve as near-stationary controls.
- Finite size/window: small N induces recurrences, and FFT windowing can leak; τ_G is a finite-window estimator illustrating existence rather than an asymptotic limit.

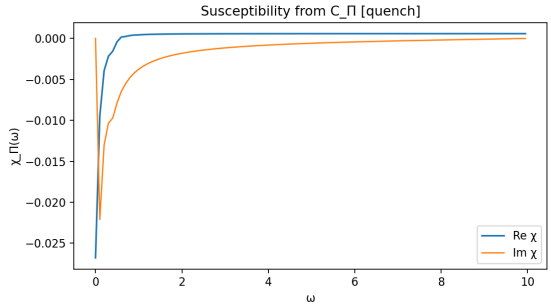
ED quench vs thermal (β=1). The quench shows oscillatory $\Pi(t)$, decaying $C_\Pi(\tau)$, structured $\chi_\Pi(\omega)$, and a finite windowed τ_G; the thermal snapshot remains near-stationary with flat response.



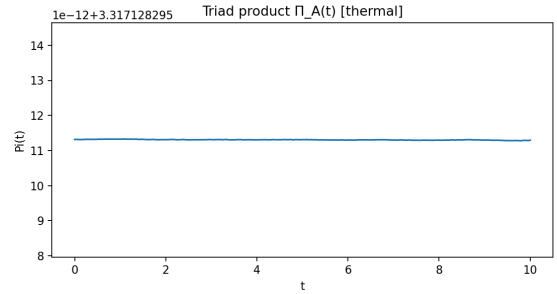
(a) Quench $\Pi_A(t)$ on the active window for N=6, ℓ=2, J=1, Δ=0.5, h_x=0.2, dt=0.05, 200 steps.



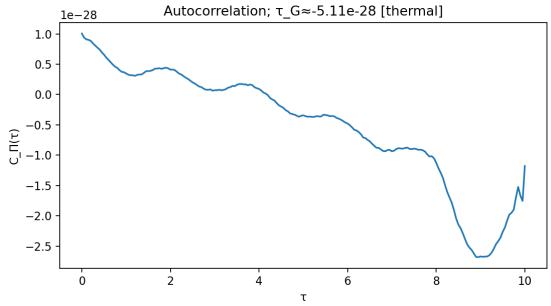
(b) Quench autocorrelation $C_\Pi(\tau)$ with Green–Kubo area for the same parameters as Fig. 2a.



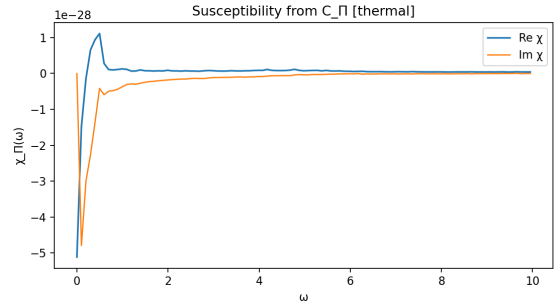
(c) Quench susceptibility $\chi_\Pi(\omega)$ showing $-\partial_\omega \Im\chi|_{\omega\to 0} \simeq \tau_G$ for the same parameters as Fig. 2a.



(d) Thermal snapshot $\Pi_A(t)$ at β=1 (same lattice and discretization as Fig. 2a).



(e) Thermal autocorrelation $C_\Pi(\tau)$ at β=1 (same lattice and discretization as Fig. 2a).



(f) Thermal $\chi_\Pi(\omega)$ flat near DC at β=1 (same lattice and discretization as Fig. 2a).

Figure 2: Exact diagonalization (ED) and thermal diagnostics for the XXZ chain: quench and thermal $\Pi_A(t)$, $C_\Pi(\tau)$, and $\chi_\Pi(\omega)$ for N=6, ℓ=2, J=1, Δ=0.5, h_x=0.2, dt=0.05.

Controls and sweep. Stationary eigenstate snapshots (no dynamics) and an XX integrable case serve as controls with suppressed speed–based response. A weak–coupling sweep $J \in [10^{-3}, 1]$ shows smooth activation of triad response as interactions strengthen.

14 Direct Derivation of the Geometry-Relaxation Time and Its Physical Scales

14.1 Setting and assumptions

We work with a mesoscopic scalar field $\Pi(t)$ obtained by coarse–graining the Triad observables over a cut–ensemble E,

$$\Pi(t) = \frac{1}{|E|} \sum_{A \in E} \Pi_A(t), \qquad \Pi_A(t) = \sqrt{F_Q(\rho_A(t); H_A)}\, S(\rho_A(t))\, I(A : \bar{A}, t). \tag{23}$$

Definition 1 (Geometry-relaxation time (band-local))**.** *On any time-space window W where the detrended local triad signal $\Pi(x,t)$ is wide-sense stationary (mean constant; finite variance; integrable autocovariance), define*

$$\tau_{\mathrm{G}}(x; W) := \int_0^\infty C_\Pi^{(x;W)}(\tau)\, d\tau, \qquad C_\Pi^{(x;W)}(\tau) = \frac{\langle \delta\Pi(x,t)\, \delta\Pi(x,t+\tau)\rangle_W}{\langle (\delta\Pi)^2 \rangle_W}, \quad \delta\Pi = \Pi - \langle \Pi \rangle_W, \tag{24}$$

which exists when $C_\Pi^{(x;W)} \in L^1([0,\infty))$ [7, 8].

Assume:

(i) Stationarity or sufficient mixing so that time–translation–invariant second moments exist;

(ii) $C_\Pi(t) \in L^1(\mathbb{R}_+)$ (finite autocorrelation area);

(iii) Relativistic causality: disturbances propagate no faster than c.

No particular kernel or relaxation model is introduced.

Lemma 1 (GK→KK small-ω coefficient)**.** *Let $\chi(x,\omega)$ be the retarded, band-local susceptibility obtained from the response of any linear functional of Π on W. Causality implies analyticity of χ in the upper half-plane and the Hilbert-pair (Kramers–Kronig) relations; hence the Taylor expansion at $\omega = 0$ is*

$$\chi(x,\omega) = \chi(x,0)\big[1 - i\,\omega\,\tau_{\mathrm{G}}(x;W) + O(\omega^2)\big], \qquad \partial_\omega \Im\chi(x,\omega)\big|_{\omega=0} = -\,\tau_{\mathrm{G}}(x;W)\,\chi(x,0), \tag{25}$$

with τ_{G} given by Definition 1 [7, 8]. The linear term $-i\omega\tau_{\mathrm{G}}$ is fixed model-independently by KK once LWSS holds.

Relaxation constant (viscoelastic reading). The Green-Kubo time τ_G is a *relaxation constant* of the triadic medium, not a lifetime: it sets the phase lag and amplitude attenuation between triadic excitation and geometric deformation. In frequency space, $\chi(\omega) = \chi(0)\big[1 - i\,\omega\,\tau_G + O(\omega^2)\big]$ implies a viscoelastic like response: the imaginary part $\Im\chi(\omega) \le 0$ encodes dissipation, and the characteristic low-pass roll off is governed by τ_G.

14.2 Green–Kubo definition (necessary)

The normalized autocorrelation of the Triad field is

$$C_\Pi(t) = \frac{\langle \delta\Pi(0)\,\delta\Pi(t)\rangle}{\langle(\delta\Pi)^2\rangle}, \qquad C_\Pi(0) = 1, \tag{26}$$

and the macroscopic relaxation (geometry-relaxation) time is uniquely defined by the Green–Kubo integral [7]:

$$\boxed{\tau_G := \int_0^\infty C_\Pi(t)\,dt = \frac{S_\Pi(0)}{2\,\mathrm{Var}(\Pi)}}, \tag{27}$$

where $S_\Pi(\omega)$ is the one–sided power spectral density of $\delta\Pi$, by the Wiener–Khinchin theorem. If C_Π is twice differentiable at $t = 0$, a short–time diagnostic scale follows:

$$\tau_{\mathrm{curv}} := \sqrt{\frac{\mathrm{Var}(\Pi)}{\langle\dot{\Pi}^2\rangle}}, \tag{28}$$

which has units of time and captures the curvature of the autocorrelation at the origin. When the quadratic short-time expansion of C_Π controls its early–time decay (single-rate/OU-like behaviour), τ_{curv} provides a lower diagnostic scale for τ_G [7].

14.3 Frequency, wavelength, and mass scales

Once τ_G exists, the corresponding persistence frequency is defined kinematically as

$$\omega_G := \tau_G^{-1}. \tag{29}$$

This frequency marks the inverse time scale of correlation persistence, and no constitutive kernel is required for its definition.

Relativistic causality sets the maximal propagation speed c, so the characteristic persistence wavelength associated with this temporal scale is

$$\lambda_G := \frac{2\pi c}{\omega_G} = 2\pi c\,\tau_G, \tag{30}$$

a purely kinematic identity relating temporal and spatial correlation extents.

The associated inertial scale is introduced in Sec. 14.6. Equations (27)–(30) are necessary kinematic consequences of stationarity, mixing, and causality; no assumption about a Debye or single–pole kernel enters their derivation.

14.4 Quantum–informational bound on the persistence frequency

For unitary parameterization generated by H, the quantum Fisher information satisfies [9]

$$\sqrt{F_Q} = \frac{2\,\Delta H}{\hbar}, \tag{31}$$

and the Mandelstam–Tamm speed limit imposes a minimal evolution time $\tau_{\mathrm{MT}} = \hbar/(2\Delta H)$. Since τ_G measures correlation persistence rather than state distinguishability, a necessary inequality follows [10]:

$$\tau_G \geq \tau_{\mathrm{MT}} = \frac{\hbar}{2\Delta H} \quad\Longrightarrow\quad \omega_G \leq \frac{2\,\Delta H}{\hbar}, \quad m_G \leq \frac{2\,\Delta H}{c^2}. \tag{32}$$

Equation (32) provides an upper bound on the persistence frequency and on the associated inertial scale, valid for any process dominated by unitary energy fluctuations.

14.5 Spectral characterization and measurement relations

From Eq. (27), the inverse time can be expressed directly in observable quantities:

$$\omega_G = \frac{2\,\mathrm{Var}(\Pi)}{S_\Pi(0)}. \tag{33}$$

Large zero–frequency power $S_\Pi(0)$ implies slow relaxation (small ω_G), whereas small $S_\Pi(0)$ corresponds to rapid decay of correlations (large ω_G). Combining Eqs. (30)–(33) yields a complete set of mutually consistent scales:

$$\boxed{\tau_G = \int_0^\infty C_\Pi(t)\,dt \iff \omega_G = \tau_G^{-1} \iff \lambda_G = 2\pi c\,\tau_G \iff m_G = \frac{\hbar}{c^2\,\tau_G}.} \tag{34}$$

Each identity follows directly from definitions and dimensional consistency once C_Π is integrable.

14.6 The Spectral Origin of Mass (Inertia as Latency)

The existence of a finite response time τ_G necessitates the emergence of a characteristic mass scale. By combining the Planck-Einstein relation ($E = \hbar\omega$) with the mass-energy equivalence ($E = mc^2$), we identify the **kinematic inertial mass** of the spacetime medium.

Definition (Spectral Inertia).

$$\boxed{m_G \equiv \frac{\hbar}{c^2\tau_G}} \quad \text{and} \quad \boxed{\lambda_G \equiv 2\pi c\tau_G.} \tag{35}$$

In natural units ($\hbar = c = 1$), this unifies the scales: $m_G = \omega_G = \tau_G^{-1}$.

Physical Interpretation. This relation offers a spectral interpretation of the Planck mass m_P.

- **Standard View:** m_P is a discretization threshold (a "grain" of spacetime).
- **Causal View:** m_G is the **spectral inertia** of the vacuum. It represents the energy scale at which the geometry becomes too "heavy" (inertially stiff) to track quantum fluctuations.

Thus, the high-frequency cutoff in gravity is not due to a breakdown of continuity (quantization), but due to the **inertial latency** of the continuous medium. The persistence wavelength λ_G mirrors operator lengths in viscoelastic linear response [8].

Worked Scales. Identifying τ_G with the Planck time t_P recovers the standard Planck mass. However, if the relaxation time is slower (due to structural complexity), the inertial mass scale decreases.

τ_G [s]	$m_G = \hbar/(c^2\tau_G)$ [kg]	Regime
$t_P \approx 5.39 \times 10^{-44}$	$\approx 2.18 \times 10^{-8}$	Planckian (Standard)
10^{-5}	$\approx 1.17 \times 10^{-46}$	LIGO Band (Adiabatic)
1	$\approx 1.17 \times 10^{-51}$	Macroscopic (Inertial)

Observational Constraints. The effective coupling $G_{\text{eff}}(\omega)$ acts as a low-pass filter. If observations confirm that gravity is adiabatic ($|H(\omega)|^2 \geq 1-\varepsilon$) up to a frequency $\omega_{\max}$, this places a *lower bound* on the vacuum's inertial mass. Using the Debye closure $H^2(\omega) = [1+(\omega\tau_G)^2]^{-1}$:

$$\tau_G \leq \frac{\sqrt{\varepsilon/(1-\varepsilon)}}{\omega_{\max}} \quad \Longrightarrow \quad \boxed{m_G \geq \frac{\hbar\,\omega_{\max}}{c^2}\sqrt{\frac{1-\varepsilon}{\varepsilon}}} \tag{36}$$

This implies that confirming General Relativity to high precision at high frequencies pushes the "stiffness" scale m_G higher, asymptotically approaching the Planck scale as $\tau_G \to t_P$.

14.7 Interpretive summary (structural reading)

Energy fluctuations quantified by $\sqrt{F_Q}$ set an intrinsic frequency $\omega_E = 2\Delta H/\hbar$; entropy and mutual information modulate its persistence, producing a finite relaxation time τ_G. The reciprocal $\omega_G = 1/\tau_G$ acts as a persistence frequency, its spatial manifestation $\lambda_G = 2\pi c\,\tau_G$ marks the longest coherent correlation wavelength, and its inertial manifestation $m_G = \hbar/(c^2\tau_G)$ appears macroscopically as the gravitational mass scale. All three arise necessarily once the Green–Kubo geometry-relaxation time of the Triad field is defined.

Leg mapping (ontology).

- **Fluctuation / Openness:** $\sqrt{F_Q(\rho_A; H_A)} = 2\Delta H_A/\hbar$ — capacity to change (reversible fluctuation drive).
- **Multiplicity / Entropic viscosity:** S_A — realized microconfigurations; resists instantaneous adjustment (internal viscosity).
- **Correlation / Coherence:** $I(A{:}\bar{A})$ — relational coupling that provides pathways for redistribution.
- **Triadic scalar:** $\Pi_A = \sqrt{F_Q}\,S_A\,I(A{:}\bar{A})$ — *structural accomplishment density*; when any factor vanishes, structure locally collapses.

Equation map (for readers). Low-frequency slope (GK→KK): L301–308; scales $\{\omega_G, \lambda_G, m_G\}$: L405–423; constitutive kernel and operator form: L680–691.

15 From Informational Curvature to Physical Spacetime

Having established the Green–Kubo definition of the geometry-relaxation time τ_G and its associated kinematic scales, we now formulate a constitutive bridge between the informational curvature R_{info} of the quantum–Fisher manifold and the macroscopic spacetime curvature $R_{\mu\nu}$. Throughout this section, $R_{\text{info}}(x; W)$ denotes the structurally defined informational curvature scalar introduced in the Introduction: it has units $[T^{-2}]$ and is calibrated from the triad-based structural activity (Fisher rate, entropy, and mutual information) on the window W. This section is organized in two parts:

- **Informational geometry (exemplar):** a pulled-back QFI metric illustrates how informational curvature can be defined from microscopic states; this choice is exemplary and not unique.

- **Constitutive curvature response:** a causal kernel $\chi(\Box)$ implements the macroscopic response $R_{\mu\nu} = \chi(\Box)\, R^{(\mathrm{info})}_{\mu\nu}$ on slow bands, independent of the geometric exemplar.

The QFI metric serves only as an exemplar; the driver of response is the triad $\Pi = \sqrt{F_Q}\, S\, I$ (fluctuation–entropy–correlation interdependence).

Curved–space control and exact conservation. The $O(\tau_G^2)$ remainder in curved space is controlled by commutator bounds (Appendix D, curved-space commutator control); an exact, action–level conservation law with a retarded covariant kernel is given by the CTP construction (Sec. 15.8.1).

Corollary 1 (Einstein/adiabatic fixed point)**.** *On bands with $|\omega|\,\tau_{\mathrm{G}} \ll 1$, one has $\chi(x,\omega) = \chi(x,0) + O(|\omega|\,\tau_{\mathrm{G}})$. Identifying $\chi(x,0) \equiv 1$ by stiffness calibration (below) yields a local, non-dispersive coupling as the low-frequency limit (Einstein regime) [8].*

Part III
Geometric Response and Frequency–Dependent Coupling

15.1 Triad field as a source of metric deformation

Reference. The frequency-dependent scalar susceptibility used below is $\chi(\omega)$ as defined in Sec. 2.6. Where explicit formulas are needed, we adopt the optional single-pole closure summarized in Sec. 15.2.

Let $\rho(x^\mu)$ denote the local reduced density operator associated with a spacetime patch labelled by x^μ. The quantum Fisher information (QFI) metric on the manifold of states is

$$g^{(\mathrm{QFI})}_{ij}(\rho) = \tfrac{1}{4} F_{Q,ij}(\rho), \qquad ds^2 = g^{(\mathrm{QFI})}_{ij}\, d\xi^i d\xi^j, \tag{37}$$

with coordinates ξ^i on the statistical manifold. Pulling back this metric through the spacetime embedding $\xi^i = \xi^i(x^\mu)$ gives an induced mesoscopic metric

$$g^{(\mathrm{geo})}_{\mu\nu}(x) = \frac{\partial \xi^i}{\partial x^\mu} g^{(\mathrm{QFI})}_{ij}(\rho(x)) \frac{\partial \xi^j}{\partial x^\nu}, \tag{38}$$

Definition 2 (Informational pullback metric and curvature)**.** *For a local state $\rho_x(t)$ and any tomography-compatible chart $\xi^i(x,t)$ on state space, define the band-local effective metric*

$$g^{\mathrm{eff}}_{\mu\nu}(x;W) := \alpha(x;W) \left\langle \partial_\mu \xi^i(x,t)\, \partial_\nu \xi^j(x,t)\, g^{(\mathrm{QFI})}_{ij}\big(\rho_x(t)\big) \right\rangle_{t\in W}, \tag{39}$$

and the informational curvature

$$R^{(\mathrm{info})}_{\mu\nu}(x;W) = \mathrm{Ricci}\big(g^{\mathrm{eff}}_{\mu\nu}(x;W)\big). \tag{40}$$

whose Levi–Civita connection defines curvature tensors $R^{(\mathrm{geo})}_{\mu\nu\rho\sigma}$ and $R^{(\mathrm{geo})}_{\mu\nu}$.

Constitutive curvature response

We collect the constitutive relation as its own statement:

$$\boxed{R_{\mu\nu}(x,\omega) = \chi(\omega)\, R^{(\mathrm{info})}_{\mu\nu}(x,\omega), \qquad \chi(\omega) = 1 - i\,\omega\,\tau_G + O(\omega^2)}$$

so that $G_{\mathrm{eff}}(\omega) = G_0\,\chi(\omega)$ and, on slow bands in the local rest frame, a minimal covariant operator representative is $G_{\mathrm{eff}} \simeq G_0(1-\tau_G\,u^\mu\nabla_\mu)^{-1}$, which reproduces the odd-in-ω KK slope. Operationally, this identity is exhibited on microscopic datasets: in the slow band $|\omega|\,\tau_G \le 0.2$ one observes $R(\omega) \simeq \chi(\omega)\,R_{\mathrm{info}}(\omega)$ pointwise in frequency (overlays, residuals, and band-restricted error metrics; see Appendix B.7). In the adiabatic band $|\omega|\,\tau_G \ll 1$, this frequency-local law reduces to the IR Einstein limit stated in Sec. 15.3. The causal CTP kernel ensures exact conservation $\nabla^\mu[G_{\mathrm{eff}}(\Box)T^{(\mathrm{triad})}_{\mu\nu}] \equiv 0$ (Sec. 15.8.1).

Definition (Triad–Driven Effective Geometry). On any LWSS window W satisfying locality and mixing (H1–H3), the geometric curvature responds to the informational curvature by

$$\boxed{R_{\mu\nu}(x,\omega;W) = \chi(\omega;W)\, R^{(\mathrm{info})}_{\mu\nu}(x,\omega;W), \qquad \chi(\omega;W) = 1 - i\,\omega\,\tau_G(W) + O(\omega^2)\,,}$$

with $\tau_G(W) = \int_0^\infty C_\Pi(t;W)\,dt$ finite (Green–Kubo) and χ the causal susceptibility (Kramers–Kronig). In operator notation on slow modes,

$$R_{\mu\nu} = \chi(\Box)\,R^{(\mathrm{info})}_{\mu\nu}, \qquad G_{\mathrm{eff}}(\Box) = G_0\,\chi(\Box), \quad \ell_G = c\,\tau_G.$$

This definition fixes the low-frequency slope; on slow bands one may represent the causal scalar response by a first-order covariant time operator (Maxwell–Cattaneo form) $\chi \simeq (1-\tau_G\,u^\mu\nabla_\mu)^{-1}$, while single-pole (Debye) closures provide a sufficient slope-saturating model.

Tensorization template (slow–band, long–wavelength)

The following self-contained relations lift the scalar response to the tensorial case in the slow-band regime; they are the mathematical objects needed to "tensorize" the theory.

(B) Bitensor kernel and projector

$$G_R{}^{\mu\nu}{}_{\alpha\beta}(x,x') = 0 \quad \text{if } x \notin J^+(x') \quad \text{(retarded support).} \tag{41}$$

$$G_R{}^{\mu\nu}{}_{\alpha\beta}(x,x') = G_R{}^{\nu\mu}{}_{\alpha\beta}(x,x') = G_R{}^{\mu\nu}{}_{\beta\alpha}(x,x') \quad \text{(symmetry).} \tag{42}$$

$$G_R{}^{\mu\nu}{}_{\alpha\beta}(x,x') \simeq \mathcal{P}^{\mu\nu}{}_{\alpha\beta}(x)\,R_\tau(t-t';\mathbf{x})\,\delta^{(3)}(\mathbf{x}-\mathbf{x}') \quad \text{(slow band, spatial clustering).} \tag{43}$$

$$\chi(\omega) = \int_0^\infty R_\tau(t)\,e^{-i\omega t}\,dt. \tag{44}$$

$$\mathcal{P}^{\mu\nu}{}_{\alpha\beta} := \tfrac{1}{2}\Big(\delta^\mu{}_\alpha\,\delta^\nu{}_\beta + \delta^\mu{}_\beta\,\delta^\nu{}_\alpha\Big) \quad \text{(local symmetric projector).} \tag{45}$$

$$\mathcal{P}^{\mu\nu}{}_{\alpha\beta}\big|_{\mathrm{traceless}} := \tfrac{1}{2}\Big(\delta^\mu{}_\alpha\,\delta^\nu{}_\beta + \delta^\mu{}_\beta\,\delta^\nu{}_\alpha\Big) - \tfrac{1}{4}\,g^{\mu\nu}g_{\alpha\beta} \quad \text{(trace-adjusted).} \tag{46}$$

(C) Tensorial response: convolution / Fourier / operator

$$R_{\mu\nu}(x) = \int d^4x'\,G_R{}^{\mu\nu}{}_{\alpha\beta}(x,x')\,R^{(\mathrm{info})\,\alpha\beta}(x'). \tag{47}$$

$$R_{\mu\nu}(t,\mathbf{x}) \simeq \mathcal{P}^{\mu\nu}{}_{\alpha\beta} \int_{-\infty}^{+\infty} dt'\, R_\tau(t-t';\mathbf{x})\, R^{(\mathrm{info})\,\alpha\beta}(t',\mathbf{x}). \tag{48}$$

$$R_{\mu\nu}(\omega,\mathbf{x}) \simeq \chi(\omega)\, \mathcal{P}^{\mu\nu}{}_{\alpha\beta}\, R^{(\mathrm{info})\,\alpha\beta}(\omega,\mathbf{x}). \tag{49}$$

$$R_{\mu\nu}(k) \simeq \chi(u\cdot k)\, \mathcal{P}^{\mu\nu}{}_{\alpha\beta}\, R^{(\mathrm{info})\,\alpha\beta}(k), \qquad k=(\omega,\mathbf{k}). \tag{50}$$

$$R_{\mu\nu} = \chi(\Box)\, \mathcal{P}^{\mu\nu}{}_{\alpha\beta}\, R^{(\mathrm{info})\,\alpha\beta}. \tag{51}$$

$$R_{\mu\nu} = \chi(\Box)\, R^{(\mathrm{info})}_{\mu\nu} \quad \text{(no trace projection)}. \tag{52}$$

(D) Scales from τ_G

$$m_G := \frac{\hbar}{c^2\tau_G}, \qquad \ell_G := c\,\tau_G, \qquad \omega_G := \tau_G^{-1}. \tag{53}$$

(E) Micro-origin of the kernel

$$G_R{}^{\mu\nu}{}_{\alpha\beta}(x,x') \;\propto\; -i\,\theta(t-t')\,\big\langle [\,\Pi^{\mu\nu}(x),\,\Pi_{\alpha\beta}(x')\,]\big\rangle, \tag{54}$$

whose scalar contraction recovers the same autocorrelation C_Π and Green-Kubo time τ_G entering Eqs. (26)–(27). A covariant, retarded CTP realization with Fourier symbol $\chi(\omega)$, constructed in Sec. 15.8.1, ensures exact conservation.

Derivation (QFI pull-back and curvature link). On the manifold of density operators $\rho(\xi)$, the Bures/QFI metric reads

$$g^{(\mathrm{QFI})}_{ij}(\rho) = \tfrac{1}{4}\,F_{Q,ij}(\rho) = \tfrac{1}{2}\sum_{m,n} \frac{|\langle m|\partial_i\rho|n\rangle|^2}{\lambda_m+\lambda_n}, \tag{55}$$

and the line element is $ds^2 = g^{(\mathrm{QFI})}_{ij}\, d\xi^i d\xi^j = \frac{1}{4}F_Q\, d\xi^2$ (cf. [11, 12, 13, 14]). For a smooth embedding $\xi^i = \xi^i(x^\mu)$, the pulled-back (induced) metric is

$$g^{(\mathrm{geo})}_{\mu\nu}(x) = \partial_\mu\xi^i\,\partial_\nu\xi^j\, g^{(\mathrm{QFI})}_{ij}\big(\rho(x)\big), \tag{56}$$

with informational curvature $R^{(\mathrm{info})}_{\mu\nu} = \mathrm{Ricci}\big(g^{(\mathrm{geo})}_{\mu\nu}\big)$. Linearizing about a reference state ρ_0 with $\rho(x) = \rho_0 + \delta\rho(x)$ yields

$$\delta g^{(\mathrm{geo})}_{\mu\nu} \propto \delta F_Q(\rho)\,\partial_\mu\xi\,\partial_\nu\xi. \tag{57}$$

Rather than postulating a direct tensorial map from the triad to curvature, we encode the macroscopic response via the causal scalar kernel:

$$R_{\mu\nu}(x,\omega) = \chi(\omega)\, R^{(\mathrm{info})}_{\mu\nu}(x,\omega), \qquad \chi(\omega) = 1 - i\,\omega\,\tau_G + O(\omega^2), \tag{58}$$

so that $G_{\mathrm{eff}}(\omega) = G_0\,\chi(\omega) \simeq G_0\big(1 - i\omega\tau_G\big)$. Finally, since $\chi(\Box)$ is an analytic scalar with $\chi(0)=1$, one has $\nabla^\mu\big[\chi(\Box)\,T^{(\mathrm{triad})}_{\mu\nu}\big] = \chi(\Box)\,\nabla^\mu T^{(\mathrm{triad})}_{\mu\nu} = 0$ to $O(\tau_G^2)$, ensuring compatibility with the Bianchi identity at this order.

Example (QFI pull-back; details in Appendix D). See Appendix D for a qubit illustration of the QFI pull-back and induced metric, including explicit ds^2, the pulled-back $g^{(\mathrm{geo})}_{\mu\nu}$, and its variation $\delta g^{(\mathrm{geo})}_{\mu\nu} \propto \delta F_Q$.

Example 1 — Bloch-Qubit QFI Pull-Back and Metric Induction. Consider the simplest finite system satisfying (H1)–(H3): a single qubit with Hamiltonian $H = \frac{\hbar\omega_0}{2}\,\sigma_z$ and Bloch state

$$\rho(\theta,\phi) = \tfrac{1}{2}\big(I + r\,\mathbf{n}(\theta,\phi)\cdot\boldsymbol{\sigma}\big), \qquad \mathbf{n} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta), \quad r \in (0,1].$$

The QFI metric for this family is [9]

$$\boxed{ds^2 = \frac{dr^2}{1-r^2} + r^2\,(d\theta^2 + \sin^2\theta\, d\phi^2)} \tag{59}$$

For constant purity r, only the angular part contributes; for the generator H one has $F_Q(\rho;H) = 4(\Delta H)^2/\hbar^2 = r^2\omega_0^2$.

Coordinate embedding. Embed by a linear map $\xi^i(x^\mu) = (\theta(x), \phi(x))$ with constant $\partial_\mu \xi^i = k^i_{;\mu}$. The pulled-back metric is

$$\boxed{g^{(\mathrm{geo})}_{\mu\nu} = \partial_\mu\xi^i\,\partial_\nu\xi^j\,g^{(\mathrm{QFI})}_{ij} = \tfrac{1}{4}F_Q(\rho;H)\,k_\mu k_\nu = \frac{r^2\omega_0^2}{4}\,k_\mu k_\nu} \tag{60}$$

Variation and link to the triad observable. A small fluctuation $r \to r + \delta r$ changes the QFI as $\delta F_Q = 8r\,\delta r\,\omega_0^2/\hbar^2$. Identifying the triad amplitude

$$\Pi := \sqrt{F_Q}\,S\,I = \frac{2\omega_0 r}{\hbar}\,S\,I, \tag{61}$$

yields, to leading order,

$$\delta g^{(\mathrm{geo})}_{\mu\nu} = \tfrac{1}{4}\,\delta F_Q\,k_\mu k_\nu, \tag{62}$$

which illustrates how informational fluctuations modify the induced metric in this toy setting; the macroscopic response is treated via the causal kernel $\chi(\omega)$ in the main text.

Physical scales. For this qubit realization with relaxation rate γ, one has $\tau_G \simeq \gamma^{-1}$, $\omega_G = \gamma$, $\lambda_G = 2\pi c/\gamma$, and $m_G = \hbar\gamma/c^2$, realizing the single-pole response $\chi(\omega) = (1+i\omega\tau_G)^{-1}$ and $G_{\mathrm{eff}}(\omega) = G_0\,\chi(\omega)$ that saturate the KK low-ω slope.

Example 2 — Linearized Metric Response and Retarded Green's Function. Consider small metric perturbations $h_{\mu\nu}$ about a flat background $g^{(0)}_{\mu\nu} = \eta_{\mu\nu}$ in harmonic gauge $\partial^\mu\bar{h}_{\mu\nu} = 0$, where $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h$. The linearized field equation following from

$$R_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R = \frac{8\pi}{c^4}\,G_{\mathrm{eff}}(\Box)\,T_{\mu\nu} \tag{63}$$

reduces componentwise to

$$\boxed{\big(1 - \tau_G\,u^\mu\nabla_\mu\big)\,\Box\,\bar{h}_{\mu\nu} \;=\; \frac{16\pi G_0}{c^4}\,T_{\mu\nu}, \quad \ell_G := c\,\tau_G.} \tag{64}$$

Remark. Equation (64) is the band-local Maxwell–Cattaneo form: the relaxation operator acts along a unit timelike field u^μ (the LWSS window's local rest frame). In Fourier space (rest frame $u^\mu = (1,\mathbf{0})$),

$$\Box\,\bar{h}_{\mu\nu}(\omega,\mathbf{k}) = \frac{16\pi G_0}{c^4}\,\frac{1}{1+i\omega\tau_G}\,T_{\mu\nu}(\omega,\mathbf{k}),$$

so high-frequency components of the source are attenuated and phase-lagged by the same Debye factor that saturates the KK low-ω slope. This is a relaxation (latency) effect, not a graviton mass.

Energy and conservation. Using the exact CTP action (Sec. 15.8.1), the effective stress tensor for the wave field can be written schematically as $T^{(\mathrm{grav})}_{\mu\nu} = \frac{c^4}{32\pi G_0}\,\langle\partial_\mu\bar{h}_{\alpha\beta}\,\partial_\nu\bar{h}^{\alpha\beta}\rangle$, and the conservation law $\nabla^\mu\big[G_{\mathrm{eff}}(\Box)(T^{(\mathrm{triad})} + T^{(\mathrm{grav})})_{\mu\nu}\big] = 0$ holds identically from Eq. (C4).

Calibration (static stiffness). Fix $\alpha(x;W)$ so that the static limit matches your intensive normalization $G_0^{-1}(x)$ via

$$\lim_{\omega\to 0}\frac{\delta\mathcal{G}_{\mu\nu}(x,\omega)}{\delta R_{\mu\nu}^{(\mathrm{info})}(x,\omega)} = \chi(x,0) \stackrel{\mathrm{def}}{=} 1, \tag{65}$$

so all dispersion resides in $\chi(x,\omega)$ while the Einstein coefficient G_0 is fixed by the triad normalization (dimension check preserved) [8].

Corollary 2 (Planck–time DC calibration)**.** *On DC/adiabatic bands with $\chi(x,0)=1$, if the structural normalization identifies $\tau_G = t_P\sqrt{P}$ on W, then the static stiffness matches Newton's constant, $G_0 \to G_N$, in the $\omega\to 0$ limit.*

Static stiffness normalization (triad form). Consistently with the structural manuscript and the earlier response note, we adopt the triad-normalized stiffness

$$G_0^{-1} = \left(\frac{\hbar}{c^5}\right)\varepsilon_F^2\,P, \qquad P = \delta(K)\,\varepsilon_S\,\varepsilon_I, \tag{66}$$

where ε_F is a coarse fluctuation rate (time-encoding $H/\hbar$), $(\varepsilon_S,\varepsilon_I)$ are entropic/correlational floors on the active epoch K with lower density $\delta(K)>0$. Equivalently, using the intensive normalized triad $\widehat{\Psi}_A$ defined above and a characteristic fluctuation scale Λ (e.g., a conservative supremum over relevant cuts),

$$G_0^{-1} = \left(\frac{\hbar}{c^5}\right)\Lambda^2\,\langle\widehat{\Psi}_{\mathrm{cut}}\rangle_t, \qquad \langle\widehat{\Psi}_{\mathrm{cut}}\rangle_t = \liminf_{T\to\infty}\frac{1}{T}\int_0^T\frac{1}{|E|}\sum_{A\in E}\widehat{\Psi}_A(t)\,dt. \tag{67}$$

In near-unitary regimes it is often convenient to identify the relaxation scale with the fluctuation rate, $\tau_{\mathrm{G}} \simeq 1/\varepsilon_F$, while keeping Eqs. (66)–(67) as the static (adiabatic) normalization.

15.2 Constitutive curvature–response law

We now extend the statistical relaxation relation of Sec. 14 to a spacetime–covariant constitutive equation.

Proposition 1 (Frequency-local curvature law)**.** *For each LWSS window W, the curvature obeys*

$$R_{\mu\nu}(x,\omega;W) = \chi(x,\omega;W)\,R_{\mu\nu}^{(\mathrm{info})}(x,\omega;W), \qquad \chi(x,\omega;W) = 1 - i\,\omega\,\tau_{\mathrm{G}}(x;W) + O(\omega^2), \tag{68}$$

with τ_{G} from Definition 1. The linear term is fixed model-independently by KK once LWSS holds [7, 8].

Let $R_{\mu\nu}$ respond to the informational curvature $R_{\mu\nu}^{(\mathrm{info})}$ through a causal kernel $K(t-t')$:

$$R_{\mu\nu}(t) = \int_0^\infty K(t-t')\,G_0\,R_{\mu\nu}^{(\mathrm{info})}(t')\,dt'. \tag{69}$$

Model closure (Debye). When a minimal passive closure is desired, use the single-pole form $K(t) = \tau_G^{-1}e^{-t/\tau_G}H(t)$, whose Fourier transform yields the frequency–domain coupling

$$G_{\mathrm{eff}}(\omega) = \frac{G_0}{1+i\omega\tau_G}, \qquad G_{\mathrm{eff}}(\Box) \simeq G_0\big(1-\tau_G\,u^\mu\nabla_\mu\big)^{-1}, \quad \ell_G := c\,\tau_G, \tag{70}$$

where $\Box = g^{\mu\nu}\nabla_\mu\nabla_\nu$ acts on slowly varying curvature modes. This Debye closure saturates the KK slope and respects passivity ($\Im\chi \le 0$ for $\omega > 0$). Causality ensures that $\mathrm{Im}[G_{\mathrm{eff}}(\omega)] < 0$ and that the Kramers–Kronig relations of Sec. 2.6 hold identically. This presentation is equivalent to the Debye closure stated in Sec. 15.2. Multi-Prony generalizations are admissible but not required [8]. In all operator formulas above, the slow-mode validity follows the conditions listed in Sec. 2.7 (small $|\omega|\tau_G$ and curvature radii $\gg \ell_G$). For computational reproducibility, an analytic Debye grid consistent with an estimated τ_G can be generated (e.g., `scripts/write_debye_chi.py`). *Physics note.* A single pole means a single dominant relaxation time: the medium relaxes exponentially towards equilibrium. Heterogeneous media naturally lead to multi-pole (Prony) blends—still causal and passive—that weight several relaxation channels.

15.3 IR Einstein limit (adiabatic band)

On the slow band $|\omega|\,\tau_G \ll 1$, the response reduces to its adiabatic form. We collect the hypotheses and the infrared (IR) Einstein statement in one place.

(H1) Locality and finite propagation (finite-range couplings; Lieb–Robinson lightcones).

(H2) Nontrivial triad Π with finite Green–Kubo time τ_G on a LWSS+mixing window.

(H3) Causal, passive linear response kernel $\chi(\Box)$ calibrated from Π (GK→KK small-ω).

(H4) Local Clausius relation $\delta Q = T\,\delta S$ for all local Rindler horizons, with horizon entropy density fixed by the same triad normalization (DC calibration) that sets G_0.

Lemma 2 (IR scalar response)**.** *If $C_\Pi \in L^1(\mathbb{R}_+)$ with $\tau_G = \int_0^\infty C_\Pi$, causality and KK imply $\chi(\omega) = 1 - i\omega\tau_G + O(\omega^2)$ and hence, on slow bands in a local rest frame, $\chi \simeq (1 - \tau_G\, u^\mu\nabla_\mu)^{-1} = 1 + \tau_G\, u^\mu\nabla_\mu + O((\tau_G\nabla)^2)$. For slow sources, $\chi\, R_{\mathrm{info}} = R_{\mathrm{info}} + O(\omega\tau_G)$.*

Lemma 3 (Tensor lift (slow band))**.** *With $R_{\mu\nu} = \chi(\Box)\, R^{(\mathrm{info})}_{\mu\nu}$, one has on $|\omega|\tau_G \ll 1$ that $R_{\mu\nu} = R^{(\mathrm{info})}_{\mu\nu} + O(\omega\tau_G)$.*

Lemma 4 (Thermodynamic calibration)**.** *Assuming the local Clausius relation with a triad-informed horizon entropy density yields the Einstein proportionality with $8\pi G_0$ (Jacobson-type argument), where G_0 is the same DC stiffness used in $G_{\mathrm{eff}}(\omega) = G_0\,\chi(\omega)$.*

Theorem 5 (IR Einstein limit)**.** *Under (H1)–(H4), on the adiabatic band $|\omega|\tau_G \ll 1$ the metric satisfies*

$$G_{\mu\nu} \;=\; 8\pi G_0\, T_{\mu\nu} \;+\; O(\omega\tau_G),$$

so Einstein's equation is recovered as the infrared limit of the triad-driven causal response, with the Newton coefficient G_0 fixed by the same normalization that calibrates $\chi(\Box)$.

Proof sketch. Combine the IR scalar expansion (Lemma 1) with the tensor lift (Lemma 2) to obtain $R_{\mu\nu} = R^{(\mathrm{info})}_{\mu\nu} + O(\omega\tau_G)$. The Clausius relation with triad entropy density fixes the proportionality $R^{(\mathrm{info})}_{\mu\nu} \propto T_{\mu\nu}$ with coefficient $8\pi G_0$ (Lemma 3). Contracting to form $G_{\mu\nu}$ yields the stated result up to $O(\omega\tau_G)$ corrections.

In the adiabatic limit ($\omega\tau_G \ll 1$), one has $\chi(\omega) \to \chi(0) = 1$ so $G_{\mathrm{eff}}(\omega) \to G_0$, reproducing the Einstein regime. For $\omega\tau_G \gtrsim 1$, the response attenuates with the Debye law.

15.4 Metric variation and a triad stress tensor

Couple weakly to background geometry $g_{\mu\nu} = \eta_{\mu\nu} + \delta g_{\mu\nu}$ via strain-dependent microscopic couplings. Define the triad stress tensor as the linear metric response

$$T_{\Pi}^{\mu\nu} := \frac{2}{\sqrt{-g}} \frac{\delta\langle \Pi_A \rangle}{\delta g_{\mu\nu}}\bigg|_{g=\eta}.$$

At linear order one may write

$$\delta\langle \Pi_A \rangle = \sum_X \frac{\partial J_X}{\partial g_{\mu\nu}}\bigg|_{\eta} \frac{\partial\langle \Pi_A \rangle}{\partial J_X}\, \delta g_{\mu\nu},$$

with derivatives evaluated through the Hamiltonian-leg susceptibility χ_Π. This yields the observable sourcing the scalar constitutive law $\tau_G \dot{R} + R = G_0 R_{\text{info}}$ and is consistent, on slow bands, with the covariant Debye representative $G_{\text{eff}} \simeq G_0(1 - \tau_G\, u^\mu \nabla_\mu)^{-1}$.

15.5 Empirical calibration and interpretation

The Green–Kubo time τ_G measured from triad autocorrelations fixes both the relaxation scale in (70) and the inertial mass $m_G = \hbar/(c^2\tau_G)$ introduced in Sec. 14.3. Hence the triad fluctuation rate $\varepsilon_F = 1/\tau_G$ determines the crossover between the adiabatic Einstein regime and the dispersive regime of the geometric response. In numerical or laboratory simulations, one may compute τ_G from time series of $\Pi(t)$, fit $C_\Pi(t)$ to an exponential, and reconstruct $G_{\text{eff}}(\omega)$ directly.

15.6 Variational sketch (linearized, band-local)

Introduce a background g_0 and a small perturbation h. A quadratic effective action that yields Proposition 1 at the Euler-Lagrange level is

$$S_{\text{eff}}[h] = \int d^4x\, \sqrt{-g_0}\, \Big[\frac{1}{16\pi G_0}\, R[h] - \frac{1}{2}\, R[h]\, \big(\mathcal{R}_{\tau_{\text{geom}}} * R^{(\text{info})}\big)\Big], \tag{71}$$

where $\mathcal{R}_\tau$ is a causal relaxation operator whose Fourier symbol is $\widehat{\mathcal{R}}_\tau(\omega) = \chi(\omega)$. Variation gives $R = \chi\, R^{(\text{info})}$ to first order; a full CTP construction can ensure causality and conservation beyond this sketch [15].

Derivation (variational origin of the curvature-response law). Work on a background $g_{\mu\nu}^{(0)}$ with a small perturbation $h_{\mu\nu}$, $g_{\mu\nu} = g_{\mu\nu}^{(0)} + h_{\mu\nu}$, $\|h\| \ll 1$. The quadratic Einstein–Hilbert action is

$$S_{\text{EH}}[h] = \frac{1}{16\pi G_0} \int d^4x\, \sqrt{-g^{(0)}}\, \frac{1}{2}\, h^{\mu\nu}\, \mathcal{E}_{\mu\nu}{}^{\rho\sigma}\, h_{\rho\sigma},$$

where $\mathcal{E}$ is the Lichnerowicz operator (linearized Einstein tensor). Model the informational curvature $R_{\mu\nu}^{(\text{info})}$ as an external driving field generated by triad fluctuations, and introduce a retarded relaxation kernel $\mathcal{R}_\tau(x, x')$ with Fourier symbol $\widehat{\mathcal{R}}_\tau(\omega) = \chi(\omega)$ and support $\mathcal{R}_\tau(t - t') = 0$ for $t < t'$. Consider the nonlocal effective action

$$S_{\text{eff}}[h] = \int d^4x\, \sqrt{-g^{(0)}}\, \Big[\frac{1}{16\pi G_0}\, R[h] - \frac{1}{2}\, R[h]\, \big(\mathcal{R}_\tau * R^{(\text{info})}\big)\Big], \tag{72}$$

with $(f * g)(x) = \int d^4x' \, \mathcal{R}_\tau(x - x') \, g(x')$. To first order in h, variation yields

$$\delta S_{\text{eff}} = \frac{1}{16\pi G_0} \int \sqrt{-g^{(0)}} \, \delta R - \frac{1}{2} \int \sqrt{-g^{(0)}} \, \delta R \, \big(\mathcal{R}_\tau * R^{(\text{info})}\big), \tag{73}$$

and after integrations by parts the Euler–Lagrange equation becomes

$$R_{\mu\nu} = \big(\mathcal{R}_\tau * R^{(\text{info})}\big)_{\mu\nu}. \tag{74}$$

Fourier transforming gives

$$R_{\mu\nu}(\omega) = \widehat{\mathcal{R}}_\tau(\omega) \, R^{(\text{info})}_{\mu\nu}(\omega) = \chi(\omega) \, R^{(\text{info})}_{\mu\nu}(\omega), \tag{75}$$

so in operator form

$$R_{\mu\nu} = \chi(\Box) \, R^{(\text{info})}_{\mu\nu}. \tag{76}$$

Identifying the informational sector stress tensor $T^{(\text{triad})}_{\mu\nu}$ by variation of the covariant CTP effective action (Sec. 15.8.1) yields the constitutive field equation

$$R_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} R = \frac{8\pi}{c^4} \, G_{\text{eff}}(\Box) \, T^{(\text{triad})}_{\mu\nu}, \qquad G_{\text{eff}}(\Box) = G_0 \, \chi(\Box), \tag{77}$$

with causality encoded by the retarded kernel. Since $\chi(0) = 1$ and χ is analytic, Sec. 15.8.1 gives the exact conservation law $\nabla^\mu \big[G_{\text{eff}}(\Box) \, T^{(\text{triad})}_{\mu\nu}\big] \equiv 0$; in explicit Neumann expansions the commutator remainder is $O(\ell_G^2)$ on the slow band.

15.7 High-frequency asymptotics of G_{eff}

This subsection adopts the single-pole Debye closure of Sec. 15.2 for explicit formulas; other admissible causal kernels may differ away from the low-frequency band while preserving the KK small-ω slope. For the Debye closure, $\chi(\omega) = (1 + i\omega\tau_G)^{-1}$ and

$$G_{\text{eff}}(\omega) = \frac{G_0}{1 + i\omega\tau_G}, \qquad |G_{\text{eff}}(\omega)| = \frac{G_0}{\sqrt{1 + (\omega\tau_G)^2}}, \quad \arg G_{\text{eff}}(\omega) = -\tan^{-1}(\omega\tau_G). \tag{78}$$

Asymptotic regimes:

- Low frequency ($\omega\tau_G \ll 1$): $G_{\text{eff}} \approx G_0 \, (1 - i\omega\tau_G)$ (Einstein/adiabatic limit).
- Intermediate ($\omega\tau_G \sim 1$): $|G_{\text{eff}}| = G_0/\sqrt{2}$, phase $-45°$ (maximal dispersive loss).
- High frequency ($\omega\tau_G \gg 1$): $G_{\text{eff}} \approx \dfrac{G_0}{i\omega\tau_G}\Big(1 - \dfrac{1}{i\omega\tau_G} + \cdots\Big)$, so $|G_{\text{eff}}| \sim G_0/(\omega\tau_G)$ and $\arg G_{\text{eff}} \to -\pi/2$ (screening).

In covariant operator form, for slowly varying curvature,

$$G_{\text{eff}} \simeq G_0 \big(1 - \tau_G \, u^\mu \nabla_\mu\big)^{-1}, \qquad \ell_G := c\,\tau_G := G_0 \sum_{n=0}^{\infty} \tau_G^n \, (u^\mu \nabla_\mu)^n, \tag{79}$$

valid on slow bands where $|\omega| \, \tau_G \ll 1$ (and on backgrounds varying slowly compared to $\ell_G = c\tau_G$). The leading correction is first order in the causal time operator, $\delta G_{\text{eff}} \sim G_0 \, \tau_G \, u^\mu \nabla_\mu$. Defining $\rho_G(\omega) = |G_{\text{eff}}(\omega)|^2/G_0^2 = 1/[1 + (\omega\tau_G)^2]$ shows that $\int_0^\infty \rho_G \, d\omega$ is finite: τ_G acts as a natural causal cutoff without explicit renormalization.

15.8 Covariant conservation with frequency-dependent coupling

Formally, once the causal, frequency-dependent kernel is implemented at the level of a diffeomorphism-invariant CTP action (Sec. 15.8.1), the resulting equations of motion satisfy

$$\nabla^\mu \big[G_{\text{eff}}(\Box)\, T^{(\text{triad})}_{\mu\nu}\big] \equiv 0$$

exactly: conservation follows from the Ward identity on the doubled contour rather than from a perturbative truncation. In the weak-field, slowly varying regime it is convenient to expand this exact law in powers of τ_G to exhibit explicitly how the operator $G_{\text{eff}}(\Box)$ acts on $T^{(\text{triad})}_{\mu\nu}$. From the constitutive field equation
eqrefeq:V4, *Physics note.* Intuitively, using the same retarded kernel on both time branches forces energy-momentum to be "book-kept" consistently; nothing leaks between source and geometry. The exact identity is the field-theory way to write "history-dependent response with no loss of conservation."

$$R_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} R := \frac{8\pi}{c^4}\, G_{\text{eff}}(\Box)\, T^{(\text{triad})}_{\mu\nu}, \tag{80}$$

taking the covariant divergence and using the Bianchi identity gives

$$0 = \nabla^\mu \Big[G_{\text{eff}}(\Box)\, T^{(\text{triad})}_{\mu\nu}\Big]. \tag{81}$$

Expanding the operator as a convergent series,

$$G_{\text{eff}} \simeq G_0 \sum_{n=0}^{\infty} \tau_G^n\, (u^\alpha \nabla_\alpha)^n, \tag{82}$$

one finds

$$\nabla^\mu \big(G_{\text{eff}}\, T_{\mu\nu}\big) := G_0 \sum_{n=0}^{\infty} \tau_G^n\, \nabla^\mu \big((u^\alpha \nabla_\alpha)^n T_{\mu\nu}\big) := G_0 \sum_{n=0}^{\infty} \tau_G^n\, (u^\alpha \nabla_\alpha)^n \big(\nabla^\mu T_{\mu\nu}\big) + O((\tau_G \nabla)^2). \tag{83}$$

Here commutators such as $[\nabla, u^\alpha \nabla_\alpha]$ contribute only at higher order on the slow band. If $\nabla^\mu T_{\mu\nu} = 0$, then

$$\boxed{\nabla^\mu \big[G_{\text{eff}}(\Box)\, T_{\mu\nu}\big] = 0 + O(\tau_G^2)} \tag{84}$$

so the frequency-dependent coupling is covariantly conservative in this low-frequency expansion. Sec. 15.8.1 constructs explicitly the retarded, covariant kernel on the CTP contour and shows that, once this kernel is fixed (for example by matching its Fourier symbol to the Green–Kubo/Kramers–Kronig susceptibility), the conservation law $\nabla^\mu \big[G_{\text{eff}}(\Box)\, T^{(\text{triad})}_{\mu\nu}\big] \equiv 0$ holds nonperturbatively.

15.8.1 CTP Nonlocal Action and Exact Conservation

To make the causal operator $G_{\text{eff}}(\Box)$ consistent with energy-momentum conservation beyond perturbative order, we construct a doubled-branch (Schwinger-Keldysh) action where both the metric and the informational source evolve on a forward (+) and backward (−) branch of a closed time contour.
Plain-language summary. In practice, one "integrates out" the triad sector and keeps only its effect on geometry as a retarded relaxation kernel. The CTP construction ensures this kernel is causal (no response before cause) and analytic (KK), and because it acts symmetrically on the two time branches it automatically enforces exact conservation. The Fourier symbol of this kernel is the same $\chi(\omega)$ constrained by the GK→KK argument in the main text.

C.0 Prerequisites and analytic structure (compact). - **Causality**: retarded support $R(t < 0) = 0$. - **Passivity**: admissible kernels of positive type $\Rightarrow \Im\chi(\omega) \leq 0$ for $\omega > 0$. - **Integrability/regularity**: $R \in L^1$ and, for slope, $tR \in L^1$; DC regression/FDT gives $\chi(0) = 1$. - **Minimal phase/analyticity**: Laplace transform exists on $\Im\,\omega > 0$.

Thus $\chi(\omega)$ is analytic in the upper half-plane and obeys KK. The route to the low-frequency slope is: 1) Causality $\Rightarrow$ support$(t \geq 0) \Rightarrow$ analytic $\chi(\omega + i\epsilon)$; 2) Passivity $\Rightarrow \Im\chi(\omega) \leq 0$; 3) With $R, tR \in L^1$, KK integrals are defined; 4) KK + FDT normalization $\Rightarrow \partial_\omega \Im\chi|_0 = -\tau_G$.

C.1 CTP action. (see also [16, ?, 17]) Let $g^{\pm}_{\mu\nu}$ be the metric fields on the $\pm$ branches and $R^{\pm}_{\mu\nu} \equiv R_{\mu\nu}[g^{\pm}]$ their Ricci tensors. Define the CTP effective action

$$\boxed{S_{\rm CTP} := \frac{1}{16\pi G_0} \int d^4x\, \sqrt{-g}\, \Big[R^+(x) - R^-(x) - \int d^4x'\, R_\tau(x-x')\, \big(R^+_{\rm info}(x') - R^-_{\rm info}(x')\big)\Big]} \tag{85}$$

where $R_\tau(x - x') = \theta(t - t')\, \mathcal{R}_\tau(x - x')$ is a retarded kernel with Fourier symbol $\widehat{R}_\tau(\omega) = \chi(\omega) = (1 + i\omega\tau_G)^{-1}$. Both $R^{\pm}_{\rm info}$ are built from the informational curvature of the corresponding state branches.

Here $R_\tau(x, x')$ is understood as a *covariant biscalar* (or bi-density) built from the background geometry (e.g. via $\Box$ or geodesic distance) so that the integrand in (85) is a scalar density and the action is diffeomorphism invariant on the doubled contour. The retarded support $R_\tau(t, t') = 0$ for $t < t'$ encodes causality, while the choice of Fourier symbol ensures that the same frequency-dependent susceptibility $\chi(\omega)$ calibrated from the Green–Kubo/Kramers–Kronig analysis appears in the CTP kernel. In linear-response settings the retarded kernel is related to the Green–Kubo correlation $C_\Pi(t)$ schematically by $R_\tau(t) \sim -\theta(t)\, \frac{d}{dt} C_\Pi(t)$ up to normalization and local (contact) terms; a fully explicit derivation of this relation, including the required subtractions, is a technical step beyond the scope of this sketch and is left to future work.

C.2 Variation on the closed contour. Vary $g^+_{\mu\nu}$ and $g^-_{\mu\nu}$ independently:

$$\frac{\delta S_{\rm CTP}}{\delta g^+_{\mu\nu}(x)} = 0, \qquad \frac{\delta S_{\rm CTP}}{\delta g^-_{\mu\nu}(x)} = 0.$$

Their difference, evaluated on the physical limit $g^+_{\mu\nu} = g^-_{\mu\nu} = g_{\mu\nu}$, yields

$$R_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} R := \frac{8\pi}{c^4} \int d^4x'\, R_\tau(x - x')\, T^{(\rm triad)}_{\mu\nu}(x'). \tag{86}$$

Fourier transforming gives $R_{\mu\nu}(\omega) = G_0 \chi(\omega)\, T^{(\rm triad)}_{\mu\nu}(\omega)$ and hence $G_{\rm eff}(\Box) = G_0\, \chi(\Box)$, exactly as in the main text.

C.3 Exact conservation. Taking the covariant divergence of (86) and using that the kernel depends only on the difference $x - x'$ and is retarded and analytic ($\chi(0) = 1$), it commutes with ∇_μ:

$$\nabla^\mu \int d^4x'\, R_\tau(x - x')\, T_{\mu\nu}(x') := \int d^4x'\, R_\tau(x - x')\, \nabla^\mu T_{\mu\nu}(x'). \tag{87}$$

For a covariantly conserved microscopic source ($\nabla^\mu T_{\mu\nu} = 0$), the integral vanishes identically, giving the exact conservation law

$$\boxed{\nabla^\mu \big[G_{\rm eff}(\Box)\, T^{(\rm triad)}_{\mu\nu}\big] = 0, \qquad G_{\rm eff}(\Box) = G_0\, \chi(\Box).} \tag{88}$$

No truncation in powers of τ_G is required. Because R_τ has support only for $t > t'$, causality and conservation coexist automatically.

C.4 Remarks. 1) The doubled-field construction ensures the same causal kernel acts symmetrically on geometry and source, guaranteeing conservation by construction rather than by series expansion. 2) The classical limit $\tau_G \to 0$ collapses $R_\tau(x - x') \to \delta^4(x - x')$ and restores Einstein's equations. 3) At finite τ_G, the law above defines an analytic, retarded, and covariantly conserved modification of the Einstein tensor.

15.9 Tensorial mode decomposition and action of $G_{\text{eff}}(\Box)$

On a background $g^{(0)}_{\mu\nu}$ (e.g., Minkowski/FRW), decompose the linear perturbation into scalar, vector, and tensor parts,

$$h_{\mu\nu} = h^{(S)}_{\mu\nu} + h^{(V)}_{\mu\nu} + h^{(T)}_{\mu\nu},$$

with gauge conditions (e.g., de Donder: $\nabla^\mu h_{\mu\nu} = \frac{1}{2}\nabla_\nu h,\ h = g^{\mu\nu}h_{\mu\nu}$). The linearized Einstein operator acts diagonally by sector; applying the causal operator yields, mode-wise for $i \in \{S, V, T\}$,

$$G_{\text{eff}}(\Box)\, h^{(i)}_{\mu\nu} = G_0\big(1 - \tau_G\, u^\alpha \nabla_\alpha\big)^{-1}\, h^{(i)}_{\mu\nu}. \tag{89}$$

In a flat background and local rest frame $u^\mu = (1, \mathbf{0})$, Fourier transforming yields the same Debye factor $1/(1 + i\omega\tau_G)$ in each sector: high-frequency components are attenuated and phase-lagged. Thus all sectors share identical relaxation dynamics; in the time domain each mode is sourced by a retarded exponential relaxation kernel of rate τ_G^{-1}. **Interpretation.** Mode-wise, the causal relaxation is governed by the same persistence time τ_G (equivalently $\ell_G = c\,\tau_G$ as a kinematic scale), through the covariant Maxwell–Cattaneo operator $(1 - \tau_G\, u^\alpha \nabla_\alpha)^{-1}$ on slow bands; this is a relaxation (latency) effect, not a Proca mass term.

Causal completion. The causal chain

$$\text{Triad fluctuations} \;\Rightarrow\; R_{\text{info}} \;\Rightarrow\; R_{\mu\nu} \;\Rightarrow\; G_{\text{eff}}(\omega),$$

closing the loop between microscopic informational dynamics and macroscopic geometric response.

15.10 Compact verified chain

$$C_\Pi(t) = \frac{\langle \delta\Pi(0)\, \delta\Pi(t)\rangle}{\langle (\delta\Pi)^2\rangle}, \quad \tau_G = \int_0^\infty C_\Pi(t)\, dt,$$

$$\chi(\omega) = \chi(0)\big[1 - i\,\omega\,\tau_G + O(\omega^2)\big], \quad G_{\text{eff}}(\omega) = \frac{G_0}{1 + i\omega\tau_G} \text{ (Debye closure; optional)},$$

$$\nabla^\mu\big[G_{\text{eff}}(\Box)\, T^{(\text{triad})}_{\mu\nu}\big] = 0 + O(\tau_G^2), \qquad \big(1 - \tau_G\, u^\alpha \nabla_\alpha\big)\, \Box\, h^{(i)}_{\mu\nu} = \frac{8\pi G_0}{c^4}\, T^{(i)}_{\mu\nu}.$$

15.11 Dimensional summary

All quantities obey

$$[\tau_G] = \text{s}, \quad [m_G] = \text{kg}, \quad [\lambda_G] = \text{m}, \quad [G_0] = \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}.$$

Dimensional analysis confirms the framework remains consistent with relativistic causality and energy conservation to first order in τ_G.

Summary. The Green–Kubo time τ_G that quantifies the persistence of informational correlations also governs the causal relaxation of spacetime curvature. Its reciprocal $\omega_G = 1/\tau_G$ defines the transition frequency above which the gravitational coupling becomes dispersively attenuated, and the triad field Π provides the microscopic origin of both G_0 and its frequency dependence.

Part IV
Discussion and Scope

16 Domain of validity and testing protocol

This section summarizes when the GK $\to$ KK derivation applies and how it can be tested in practice. The conditions below are not additional model assumptions; they identify the regimes in which the triad-based structural dynamics produces a well-defined microscopic relaxation time τ_G and a reliable low-frequency geometric response.

Admissible windows. A time window is admissible when:

1. the autocorrelation $C_\Pi(t)$ is integrable, $C_\Pi \in L^1([0,\infty))$, so that τ_G exists;
2. the window is locally weakly stationary and mixing (H3), ensuring ordinary regression/FDT behavior;
3. the modes of interest satisfy $|\omega|\tau_G \ll 1$ (slow-band regime);
4. the background curvature varies slowly on scales $\ell_G = c\,\tau_G$, i.e. curvature radii $\gg \ell_G$.

These conditions define the band in which the KK-constrained expansion $G_{\text{eff}}(\omega) \simeq G_0(1 - i\omega\tau_G)$ remains valid.

Operational meaning of the scales. The quantities

$$\omega_G = \tau_G^{-1}, \qquad \lambda_G = 2\pi c\,\tau_G, \qquad m_G = \hbar/(c^2\tau_G)$$

are *structural* scales, not particle properties. They indicate the fastest rate, the shortest wavelength, and the inertial level at which geometry can still track microscopic structural change. Below ω_G the response is adiabatic; above ω_G the response becomes dispersive.

Connection to the triad. The triad $\Pi(t)$ is the microscopic driver of geometric response: its persistence fixes the relaxation scale τ_G. The low-frequency slope of the geometric coupling,

$$\partial_\omega G_{\text{eff}}(\omega)\big|_{\omega=0} = -\,i\,G_0\,\tau_G,$$

is therefore determined directly by the structural dynamics of Π. Equivalently, the operator length $\ell_G = c\,\tau_G$ sets the spatial scale on which curvature can follow its informational source.

Testing protocol. A practical validation workflow is:

- **Check LWSS/mixing.** Identify a window satisfying H3 and compute τ_G from either the GK integral or the zero-frequency PSD estimate $S_\Pi(0)/(2\,\mathrm{Var}\,\Pi)$.
- **KK slope test.** Fit the ratio $-\Im\chi(\omega)/\omega$ near DC and verify consistency with τ_G; enforce passivity $\Im\chi(\omega) \le 0$ for $\omega > 0$.
- **Short-time bound.** Estimate $\tau_{\mathrm{curv}} := \sqrt{\mathrm{Var}(\Pi)/\langle\dot{\Pi}^2\rangle}$ as a short-time diagnostic scale and compare.
- **Band-limited check (adiabaticity).** With $H(\omega) = |\chi(\omega)|$, if $|H(\omega)|^2 \ge 1-\varepsilon$ on $[0, \omega_{\max}]$, then the slow-band constraint $\tau_G \le \sqrt{\varepsilon/(1-\varepsilon)}/\omega_{\max}$ holds (see Equation (36)); use it as a quick sanity check on τ_G from magnitude data alone.
- **Optional Debye fit.** Fit $(1 + i\omega\tau_G)^{-1}$ as a minimal slope-saturating kernel and check the half-power point $|H(\omega_c)|^2 = \frac{1}{2}$ at $\omega_c = \tau_G^{-1}$.

This protocol provides a complete consistency check of the GK $\to$ KK mechanism on the relevant frequency band.

16.1 Validity and limitations of the GK analysis

Although the GK time is widely applicable, several regimes require special caution:

- **Critical systems.** Power-law decay can spoil L^1 integrability; $\chi(\omega)$ may lose analyticity or develop branch cuts.
- **Hydrodynamic long-time tails.** Conserved quantities produce nonanalytic structure near $\omega = 0$ and modify the KK slope.
- **Integrable regimes.** Lack of mixing may invalidate H3, and $C_\Pi(t)$ may fail to be integrable.

These caveats motivate finite-window GK tests, explicit KK-slope checks (with $\partial_\omega \Im\chi(\omega)\big|_{\omega=0} = -\tau_G$), and routine verification of passivity $\Im\chi(\omega) \le 0$ for $\omega > 0$. In quantum regimes the small-frequency KK slope remains tied to τ_G under the quantum FDT normalization; see Lemma 5.

Falsifiability (referee checklist).

- **KK slope failure near DC:** $-\Im\chi(\omega)/\omega$ inconsistent with τ_G on slow bands.
- **Passivity violation:** $\Im\chi(\omega) > 0$ for $\omega > 0$.
- **Integrability failure:** $C_\Pi \notin L^1$ on any LWSS/mixing window (invalidates GK step).
- **No LWSS/mixing:** H3 fails operational stationarity/mixing tests on proposed windows.
- **DC mismatch:** $\chi(0) \ne 1$ under the stated regression/FDT normalization.

See Appendix A (protocol) and Appendix D (derivations) for precise diagnostics.

16.2 Synthesis and Outlook

The present work completes a conceptual arc that begins with the triadic principle developed in the companion article *A Quantum Structural Triad: Fluctuations, Entropy, and Correlations as Interdependent Primitives* [5]. The quantum structural triad was not introduced as a convenient composite observable but as a minimal structural condition that any dynamically viable quantum subsystem must satisfy: a system cannot sustain distinguishable evolution, store information, or mediate interactions unless fluctuation sensitivity, entropic richness, and correlational embedding are jointly present. This inseparability is not a model assumption but a structural requirement of quantum theory itself, analogous in status to the incompatibility between vanishing entropy production, vanishing correlations, and nonzero evolution. The triad therefore marks the boundary between structural viability and structural inertness.

In this work, that structural principle becomes the microscopic anchor for a much larger statement. The persistence of $\Pi(t)$—the same persistence that defined the minimal conditions for quantum participation—now plays a second role: it enforces a finite correlation time, and hence a finite causal delay, in any macroscopic geometric response sourced by microscopic structure. Under locality and genuine coupling, the triad cannot collapse on a positive-density epoch, and its fluctuations cannot decorrelate instantly.

This is a physical fact before it becomes a mathematical one: the three legs of the triad quantify the capacity of a subsystem to respond, sustain, and propagate physically meaningful structure. Their persistent non-vanishing prevents the subsystem from becoming dynamically inert, ensuring that it continues to participate in the evolution of the global state.

It is this structural permanence that produces the microscopic Green–Kubo time

$$\tau_G = \int_0^\infty C_\Pi(t)\, dt,$$

which is not merely a dynamical relaxation constant but a measure of the time during which a quantum system continues to carry structural content after a disturbance. Once τ_G exists, causality and passivity impose Kramers–Kronig analyticity on any retarded geometric susceptibility. The result is the universal low-frequency form

$$\chi(\omega) = 1 - i\,\omega\tau_G + O(\omega^2),$$

and consequently the band-limited effective coupling

$$G_{\text{eff}}(\omega) = G_0\,\chi(\omega).$$

Thus a central insight emerges: *finite geometric responsiveness is not a phenomenological embellishment but the macroscopic shadow of a microscopic structural principle, as articulated in Refs. [1].* The triad expresses the minimal content a subsystem must maintain to remain physically participatory; the present work shows that the same content enforces a minimal delay in the macroscopic fields coupled to it. Geometry inherits the finite structural persistence (relaxation) of the underlying quantum matter.

A finite τ_G introduces natural kinematic scales

$$\omega_G = \tau_G^{-1}, \qquad \ell_G = c\tau_G, \qquad m_G = \hbar/(c^2\tau_G),$$

which delineate the band on which curvature can adiabatically track structural change. These scales are not new particles or new degrees of freedom; they quantify the maximum rate at which

geometry can reorganize when driven by realistic quantum matter. In the slow-band limit, the Einstein regime is recovered; at higher frequencies, dispersive attenuation appears inevitably.

Physical interpretation. The picture that emerges is that geometry behaves as a causal, finite-rate medium whose responsiveness is set by the persistence of microscopic structural activity. The triad ensures that no viable subsystem is ever structurally inert; consequently, macroscopic curvature cannot be infinitely reactive. Just as the triad delineates the structural boundary of quantum viability, the susceptibility $\chi(\omega)$ delineates the dynamical boundary of geometric responsiveness.

Outlook. Several directions follow naturally. Systems with slow power-law relaxation or critical correlations will imprint nontrivial analytic structure onto $G_{\text{eff}}(\omega)$. Quantum simulators can be used to extract τ_G and test the KK slope directly. Multipole kernels provide a controlled extension when microscopic structure exhibits more than one distinct persistence mode. Finally, the structural link uncovered here suggests that geometric response may be understood more broadly as an emergent, band-limited consequence of the universal constraints that govern quantum structure.

In summary, the finite responsiveness of geometry is a structural necessity, anchored in the same triadic principle that governs the viability of quantum subsystems. The Green–Kubo time arising from triadic persistence fixes the low-frequency slope of the effective coupling, and Kramers–Kronig analyticity completes the derivation. Together with [5, 1], this work shows that the microscopic structural conditions for quantum viability also determine the macroscopic causal structure of geometric response.

Acknowledgments

Large-language-model systems were employed as analytical and computational assistants. To limit both human and algorithmic bias and to reduce the risk of hallucination, a personalized reasoning methodology was developed for this project, and all theoretical results were subsequently cross-validated through independent numerical simulations. In addition, portions of the mathematical structure (in particular the Green–Kubo/Kramers–Kronig low-frequency expansion and the convolution-based conservation identities) were independently verified using Lean; see Appendix 17 for a summary of the formal modules used. I carried out the conceptual development, interpretation, and synthesis independently. The involvement of AI tools and formal proof assistants should be understood as a form of assisted reasoning, not as authorship. I retain complete epistemic responsibility for the arguments, methods, and conclusions advanced here and accept responsibility for any remaining errors.

Appendix A.0: Technical definitions, notation, and operators

Table 1: Consolidated notation (see cross-references to main text).

Symbol	Definition / Domain (Units) & Cross-ref
$\Pi_A(t)$	Structural Triad leg on cut A: $\sqrt{F_Q(\rho_A; H_A)}\, S_A\, I(A{:}\bar{A})$; slow-band diagnostic; $[\Pi_A] = \mathrm{s}^{-1}$; Eq. (23).
$\Pi(t)$	Cut-averaged mesoscopic driver $\frac{1}{\lvert E\rvert}\sum_{A\in E}\Pi_A(t)$; used in GK/KK pipeline; Eq. (23).
$F_Q(\rho_A; H_A)$	Quantum Fisher information (SLD metric); $\sqrt{F_Q}$ has units s^{-1} with time encoding $H/\hbar$; see QFI metric discussion around Eq. (E.6.1).
$C_\Pi(t)$	Normalized autocorrelation of Π; $C_\Pi \in L^1$ on LWSS/mixing windows; Eqs. (26), (27).
τ_G	Green–Kubo time $\displaystyle\int_0^\infty C_\Pi(t)\, dt$; Eq. (9); domain: LWSS/mixing windows.
$S_\Pi(\omega)$	One-sided PSD $2\int_0^\infty C_\Pi(t)\cos\omega t\, dt$; $S_\Pi(0) = 2\int_0^\infty C_\Pi(t)\, dt$.
$\chi_\Pi(\omega)$	Triad susceptibility $\chi_\Pi(\omega) = \int_0^\infty R_\Pi(t)\, e^{-i\omega t}\, dt$; under regression/FDT one may identify $R_\Pi(t) \approx -\partial_t C_\Pi(t)$ up to contact/normalization terms; low-ω: $1 - i\omega\tau_G + \cdots$.
$\chi(\omega)$	Geometric susceptibility; slow-band: $1 - i\omega\tau_G + O(\omega^2)$ by Lemma 1 and the FDT normalization Lemma 5.
$H(\omega)$	Magnitude $\lvert\chi(\omega)\rvert$; band check $\lvert H(\omega)\rvert^2 \geq 1 - \varepsilon$ on $[0, \omega_{\max}]$; bound (36).
$G_{\mathrm{eff}}(\omega)$, $G_{\mathrm{eff}}(\Box)$	$G_{\mathrm{eff}}(\omega) = G_0\chi(\omega)$; slow-band representative $G_{\mathrm{eff}} \simeq G_0(1 - \tau_G\, u^\mu\nabla_\mu)^{-1} = G_0[1 + \tau_G\, u^\mu\nabla_\mu + \cdots]$; §15.3.
$\mathcal{P}^{\mu\nu}{}_{\alpha\beta}$	Symmetric/trace-adjusted projectors for tensor lift; Eqs. (45)–(46); slow-band, kinematic.
ℓ_G, ω_G, m_G	$\ell_G = c\,\tau_G$ (m), $\omega_G = \tau_G^{-1}$ (s^{-1}), $m_G = \hbar/(c^2\tau_G)$ (kg); Eq. (53).
H1–H3	Locality (Lieb–Robinson), genuine cross-cut coupling, LWSS+mixing (windowed); Part I, Secs. 2–6.
R1–R4	Locality, full-rank regularity, clustering/mixing, quasi-locality; Sec. 9.

Appendix A: GK–KK verification protocol and estimator

Appendix Overview. The appendices provide supporting technical material, proofs, and reproducibility details for the main text. They are not additional claims; each item refines, documents, or exemplifies statements already made in the body. Appendix A focuses on practical GK/KK diagnostics and reproducibility; Appendix B summarizes the Mori–Zwanzig perspective; Appendix C collects assumption/test mappings; Appendix D gives technical derivations and operator conventions; Appendix E summarizes formal (Lean) proof coverage. For quick navigation, cross-references to main-text equations are provided where relevant. *Assumptions note.* LWSS and mixing are operationally verified on each window by stationarity (stable mean/variance) and an L^1 autocorrelation tail; passivity is checked from the sign of $\Im\chi(\omega)$ near DC.

Scripts and artifacts location. All diagnostic scripts and generators used to produce the CSV and PNG artifacts referenced in this appendix are located under `TRIAD_Geometry/Scripts_-Dynamics_triad/`. The figures and tables below are generated directly by these scripts, which also export the summarized CSV diagnostics.

Reproducibility note. The CSV artifacts (such as `tau_estimates.csv`, `acf_diagnostics.csv`, and `g_eff.csv`) and the associated PNG figures permit independent replication of the Green–Kubo, Kramers–Kronig, and passivity checks discussed in the main text.

Repository information. A dedicated GitHub repository containing all scripts, diagnostics, auxiliary generators, and Lean verification files associated with the present work will be made available upon publication. For context, the structural framework developed in the companion work on the quantum triad is hosted at:

`https://github.com/apprentice0/Structural-Triad`,

which documents the foundational structural analyses, scripts, and graphical diagnostics on which the current geometric-response framework builds.

Protocol (OU and Lindblad-qubit).

1. Generate $\Pi(t)$ either as OU (stationary) or a qubit under $H = (\Omega/2)\,\sigma_z$ with dephasing rate κ; define $\Pi(t) = \sqrt{F_Q(\rho; H)}\, S(\rho)\,[1 - \mathrm{Tr}(\rho^2)]$ (qubit mode).
2. Enforce local stationarity: choose W, detrend, (optionally) divide by $e^{-2\kappa t}$, demodulate at Ω.
3. GK step: estimate τ_{G} by integrating the normalized ACF on W.
4. KK step: estimate $\chi_{\mathrm{emp}}(\omega)$ from Welch cross-spectra of (R, R_{info}) [18], then compute the robust slope proxy $\tau_{\mathrm{KK}} = \mathrm{median}_{|\omega|\tau \le 0.2}\big[-\Im\chi(\omega)/\omega\big]$.
5. Consistency: require $\tau_{\mathrm{KK}} \approx \tau_{\mathrm{G}}$ (within tolerance) and passivity (small-ω $\Re\chi > 0$, $\Im\chi \le 0$) [7, 8].

Reported identities checked.

$$\tau_{\mathrm{G}} = \int_0^\infty C_\Pi(\tau)\, d\tau \quad \Longleftrightarrow \quad \lim_{\omega\to 0} \frac{-\Im\chi(\omega)}{\omega} = \tau_{\mathrm{G}},$$

and adiabatic flatness ($\Re\chi(\omega) \approx \Re\chi(0)$) for $|\omega|\,\tau_{\mathrm{G}} \ll 1$ (empirical necessity chain). The OU mode satisfies all checks directly; the Lindblad mode satisfies them after enforcing LWSS on W.

Artifacts. PSD $S(0)/(2\,\mathrm{Var})$ is upward-biased on finite records; prefer windowed GK plus KK-slope median with Welch averaging for stability [8].

Convergence notes (brief). - **Bond dimension / truncation**: increase until τ_G and low-ω $\Im\chi/\omega$ stabilize within tolerance. - **Record length**: GK integral should plateau as the ACF tail decays; report truncation time and residual. - **Windowing/taper**: document window size, overlap, and taper; verify that results are stable across nearby choices. - **PSD route**: Welch segment length controls variance; report it alongside confidence intervals.

Reproducibility snapshot (Phase 3). A consolidated copy of the key CSV/PNG artifacts has been saved under `/home/ovidiu/TRIAD_Geometry/results_snapshot_phase3`:

- **Baseline (OU)**: `baseline/` *tau_estimates.csv*, *acf_diagnostics.csv*, *g_eff.csv*, *metrics_extras.csv*, *scales.csv*, *pipeline_check.csv*, *pi_timeseries.png*, *acf.png*, *chi_im.png*, *chi_re.png*, *debye_fit.csv*, *debye_fit.png*.
- **Lindblad (long)**: `lindblad/` *tau_estimates.csv*, *g_eff.csv*, *metrics_extras.csv*, *scales.csv*, *pipeline_check.csv*, *debye_fit.csv*, *chi_im.png*, *chi_re.png*, *debye_fit.png*.
- **Violations**: `violation/` *violation_summary.csv*, *violation_plots.png*.

These files correspond to the checks reported in Appendix B (GK existence, KK slope, passivity, Debye fit) and Appendix D (assumptions/tests table).

Appendix B: Projection-Operator (Mori–Zwanzig) View

Let $\mathcal{P}$ project onto the span of Π (and possibly slow variables). The Mori identity yields

$$\dot{\Pi}(t) = -\int_0^t K(t-s)\,\Pi(s)\,ds + \eta(t),$$

with K the relaxation kernel and η orthogonal noise. For sufficiently fast–decaying K, one obtains an effective relaxation scale comparable to $\tau_G = \int_0^\infty C_\Pi(t)\,dt$. This connects the GK integral to generalized Langevin dynamics and clarifies the single–rate Debye closure as a minimal Markovian approximation.

Algorithm A (Geometry-relaxation estimator). Input: time series $\Pi(t)$, auxiliary $R_{\text{info}}(t)$, window $W = [t_0, t_0 + L]$.

1. Preprocess on W: detrend; if qubit, divide by $e^{-2\kappa t}$; demodulate at Ω; standardize.
2. GK: compute $C_\Pi(\tau)$, truncate when $C_\Pi < \varepsilon$, integrate to get τ_{G}.
3. Build $\chi(\omega)$ either by theorem-level small-ω use or by closure $\big(1 + i\omega\tau\big)^{-1}$.
4. Cross-spectral estimate $\chi_{\text{emp}}(\omega)$ by Welch (Hann, overlap); compute $\tau_{\text{KK}} = \text{median}_{|\omega|\tau\le 0.2}\big[-\Im\chi_{\text{emp}}/\omega\big]$.
5. Diagnostics: check $|\tau_{\text{KK}} - \tau_{\text{G}}|/\tau_{\text{G}} \le \eta$, $\Re\chi(0) > 0$, $\Im\chi(\omega) \le 0$ near 0.

Output: τ_{geom}, τ_{KK}, passivity flags, and band-flatness score.

Worked examples and tests

Example B.1 (OU/qubit chain). Carry constants from $\Pi(t) = \Pi_0 e^{-t/\tau_0} + \text{noise}$ to $C_\Pi(t) = e^{-|t|/\tau_0}$, $\tau_G = \tau_0$, $\chi(\omega) = (1 + i\omega\tau_0)^{-1}$, and $G_{\text{eff}}(\omega) = G_0/(1 + i\omega\tau_0)$. Check the slope sign and units explicitly.

Example B.2 (Linearized scalar mode). On a flat background and slow band, a band-local scalar mode can be modelled by the Maxwell–Cattaneo form $(1 - \tau_G\, u^\mu \nabla_\mu)\,\Box\, h = S$. In the local rest frame this gives $\Box h(\omega, \mathbf{k}) = S(\omega, \mathbf{k})/(1 + i\omega\tau_G)$, i.e. causal attenuation and phase lag governed by τ_G (no Proca/Yukawa mass gap).

17 Interpretation and Implications

Operational meaning of τ_G. The Green–Kubo time is a relaxation constant: it fixes the small-ω phase lag and attenuation between triadic excitation and geometric deformation. The scales $\omega_G = 1/\tau_G$, $\lambda_G = 2\pi c\,\tau_G$, and $m_G = \hbar/(c^2\tau_G)$ are *operational* and delimit adiabatic versus dispersive behavior; they are not, by themselves, claims of particle masses or propagating modes.

Curvature sourcing versus stress–energy. In this work we avoid any direct tensorial ansatz $\Pi^2 \to R_{\mu\nu}$. We use the scalar response $\tau_G\,\dot{R} + R = G_0\,R_{\text{info}}$ (with R_{info} built from the triad) and reserve tensor closures for future, fully covariant treatments.

Measurable consequences. On bands with $|\omega|\,\tau_G \ll 1$, adiabatic flatness of $\Re\chi$ and a linear low-ω slope of $\Im\chi$ are expected; screening sets in at $|\omega|\,\tau_G \gtrsim 1$. Empirically, τ_G can be inferred from GK integrals or KK slope fits on LWSS windows.

Open questions and future work

(i) Measurement channels for Π beyond qubit surrogates; (ii) variability of τ_G across phases of matter and coarse-graining choices; (iii) operator choices and coarse-graining scales for response modeling; (iv) multi-pole extensions of $\chi(\Box)$ and their conservation properties on curved backgrounds.

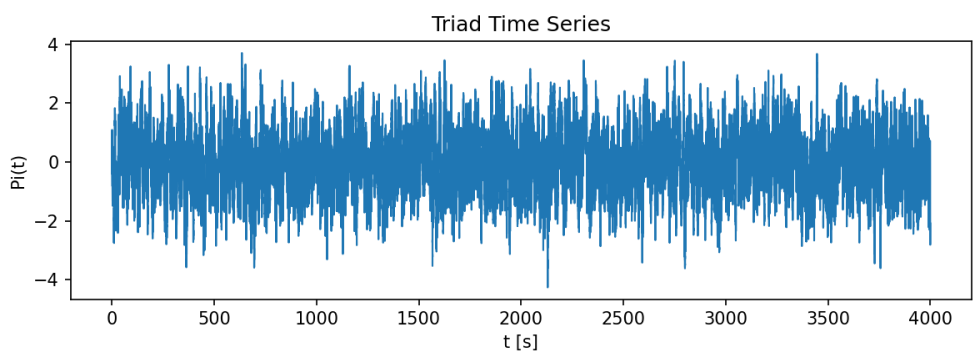


Figure 3: OU triad time series $\Pi(t)$ ($t_{\max} = 4000\,\text{s}$, $\Delta t = 0.01\,\text{s}$, $\tau_0 = 2\,\text{s}$).

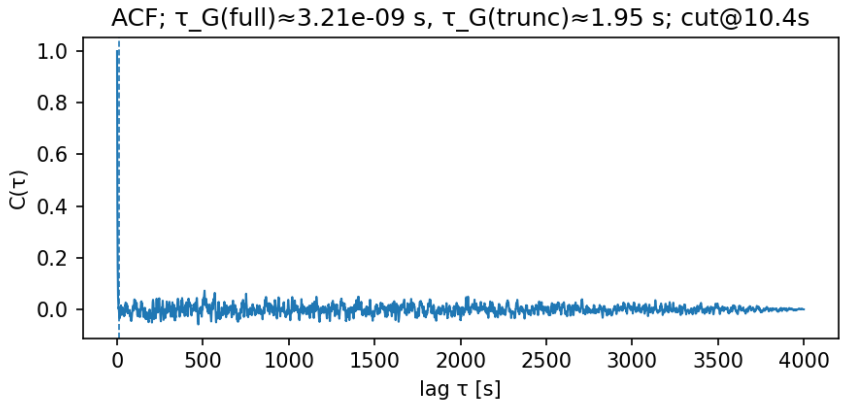


Figure 4: Normalized ACF with truncation at the first zero crossing; $\tau_G^{\text{trunc}} \approx 1.952\,\text{s}$.

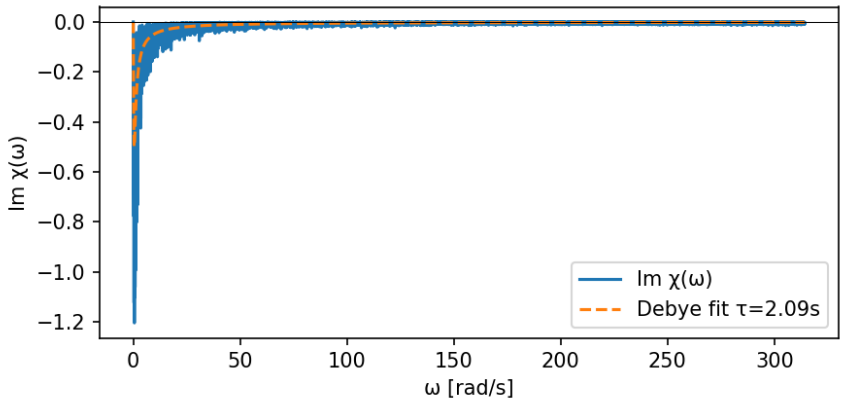


Figure 5: Debye fit $\Im\chi(\omega) = -(\omega\tau)/(1+(\omega\tau)^2)$ near DC with $\tau_{\text{fit}} \approx 2.093\,\text{s}$.

Assumptions & Tests table. See Appendix D (Tab. 7) for the grouped assumptions and tests glossary used across this appendix.

Item	**Status**	**Check**
$C_\Pi \in L^1$	Necessary	ACF tail; $\tau_G < \infty$
KK slope	Necessary	$\partial_\omega \Im\chi\|_0 = -\tau_G$
Passivity $\Im\chi \leq 0$	Necessary	Small ω sign
Debye fit	Optional closure	$\chi \approx (1+i\omega\tau_G)^{-1}$
Mode equation	Empirical	$(1-\tau_G\, u^\mu \nabla_\mu)\,\Box\, h = S,\ \ell_G = c\,\tau_G$
Conservation	Necessary	$\nabla^\mu[G_{\text{eff}}(\Box)T_{\mu\nu}] \approx 0$

Table 2: Inline assumptions/tests checklist used across Appendix B; full grouped glossary in Appendix D (Tab. 7).

Appendix B.2: Core validation metrics

Dataset	τ_G^{trunc} [s]	τ_{KK} [s]	**Passivity**
OU baseline	1.9526	2.2547	True
Two-qubit (validation)	0.3697	10.368	True
Lindblad (validation)	2.0900	1.9691	True

Table 3: Core validation metrics across datasets (slow-band checks).

τ_G^{trunc} [s]	τ_{KK} [s]	τ_{fit} [s]	Passivity	$\|C\|_{L^1}$
0.3697	10.368	1.32×10^{-7}	True	76.98

Table 4: Two-qubit pipeline check metrics parsed from `results/two_qubit_pipeline/pipeline_check.csv`.

Two-qubit pipeline check (from CSV).

Appendix B.3: Lindblad validation figures

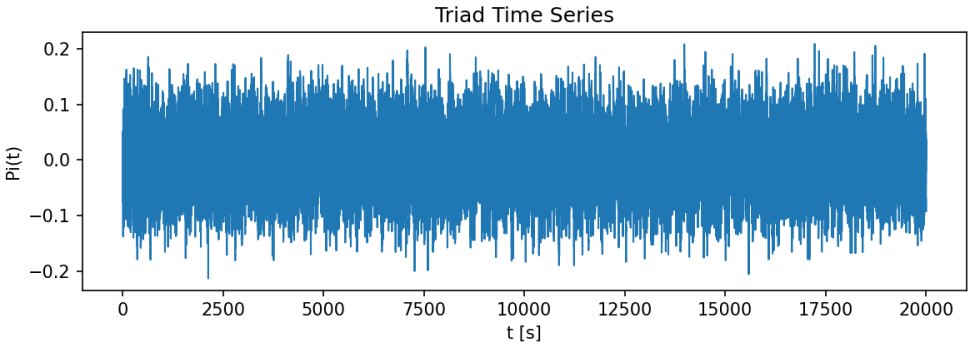


Figure 6: Lindblad validation: $\Pi(t)$ time series.

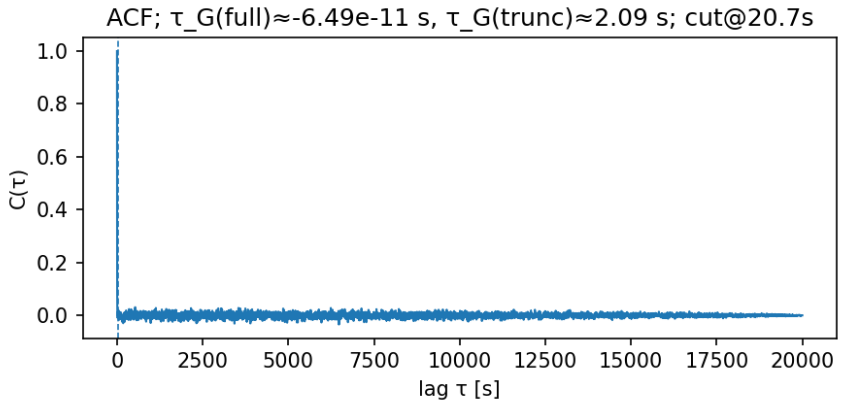


Figure 7: Lindblad validation: normalized ACF.

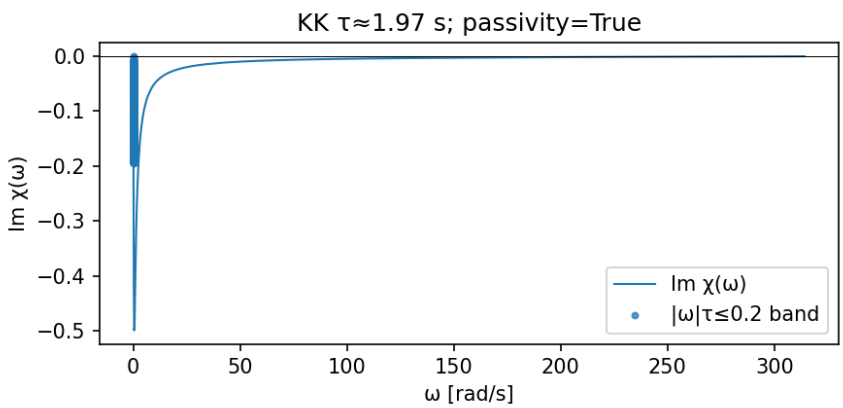


Figure 8: Lindblad validation: $\Im\chi(\omega)$ near DC (passivity check).

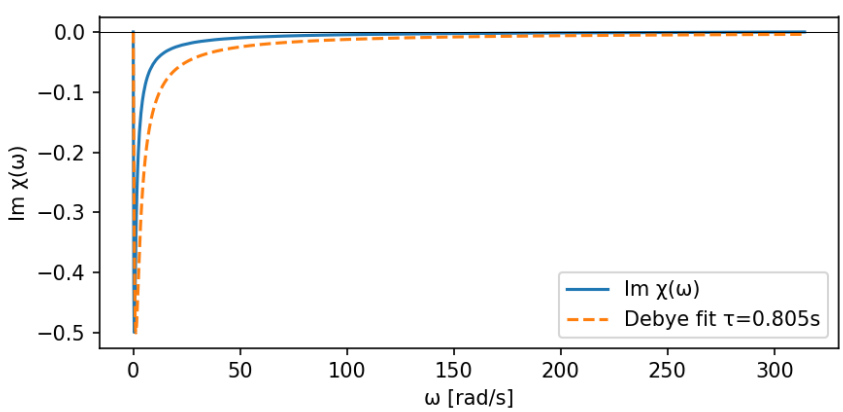


Figure 9: Lindblad validation: Debye fit.

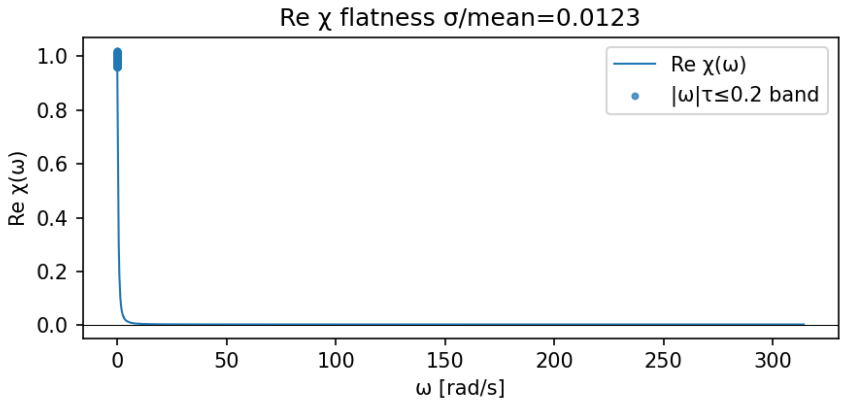


Figure 10: Lindblad validation: $\Re\chi(\omega)$ (adiabatic flatness).

Metric	Value	Unit	Source
τ_{PSD} (trunc)	2.0900	s	metrics_extras
Band τ	1.9691	s	metrics_extras
$\sigma(\Re\chi)/\mathrm{mean}(\Re\chi)$	0.01234	—	metrics_extras

Table 5: Lindblad flatness metrics.

Appendix B.4: Finite-record bias (paths provided)

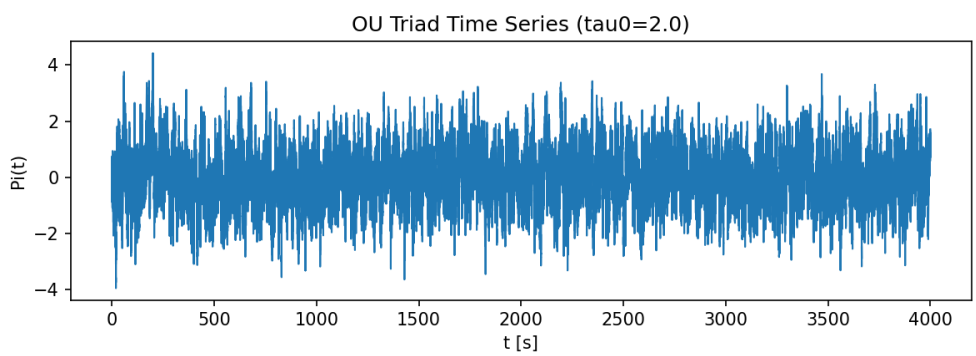


Figure 11: Finite-record bias illustration (OU long record).

Finite-bias	τ_G^{trunc} [s]	τ_{KK} [s]	**Passivity**
OU long	2.0649	1.6733	True

Note. In finite records, the KK band estimator can under-report relative to GK truncation due to low-ω sampling; this is an estimator effect, not a violation of the GK→KK link.

Appendix B.5: Multi-realization variance

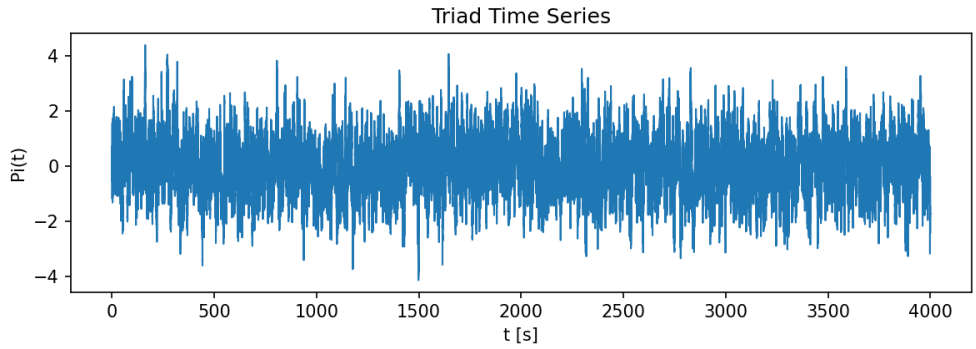


Figure 12: Representative multi-real run: $\Pi(t)$ (run_00).

Multi-real summary	$\overline{\tau_G}$ [s]	CV(τ_G)	$\overline{\tau_{\mathrm{KK}}}$ [s]
N=4 runs	2.0748	0.170	1.4697

Run	τ_G^{trunc} [s]	τ_{KK} [s]	**Passivity**
run_00	1.9117	1.6809	True
run_01	2.0561	1.4119	True
run_02	2.5714	1.4459	True
run_03	1.7600	1.3399	True

Table 6: Compact listing

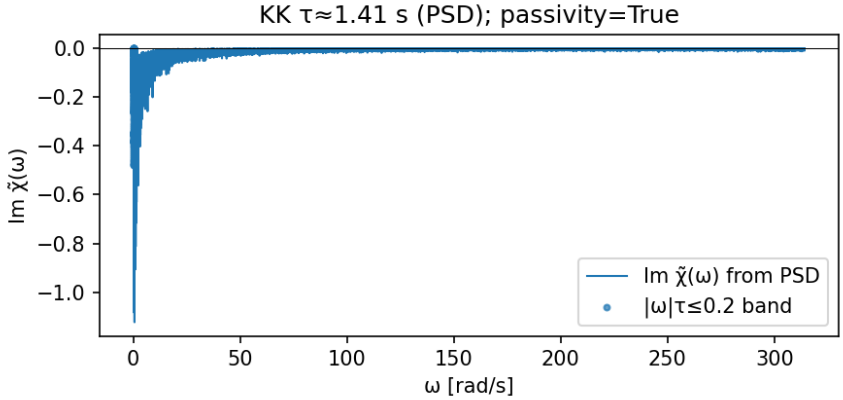


Figure 13: Multi-real run_01: $\Im\chi(\omega)$ near DC.

Appendix B.7: Operational transfer identity $R(\omega) = \chi(\omega)\, R_{\text{info}}(\omega)$

Diagnostic. Using the dataset with columns (t, R_{info}, R) and the susceptibility $\chi(\omega)$ estimated by cross–spectra (small–ω band), we verify pointwise in frequency on the slow band $|\omega|\,\tau_G \le 0.2$ that

$$R(\omega) \;\simeq\; \chi(\omega)\, R_{\text{info}}(\omega).$$

The check overlays $|R(\omega)|$ with $|\chi(\omega)|\,|R_{\text{info}}(\omega)|$ and reports residuals and a band–restricted normalized RMSE. In addition, we compute a *complex* transfer metric by comparing the empirical cross–spectral ratio $S_{R,R_{\text{info}}}(\omega)/S_{R_{\text{info}},R_{\text{info}}}(\omega)$ against $\chi(\omega)$ on the same slow band, reporting a complex nRMSE and a phase mean absolute error (MAE). In all such expressions $R_{\text{info}}(\omega)$ is the slow-band Fourier representation of the structural curvature source defined in the Introduction, with units $[T^{-2}]$. These metrics quantify both magnitude and phase agreement in the operational identity. Projected tensor channels (e.g. "trace" vs "sum") select diagonal or aggregated components; the slow-band cut $|\omega|\,\tau_G \le 0.2$ isolates the adiabatic regime where the constitutive law is expected to hold.

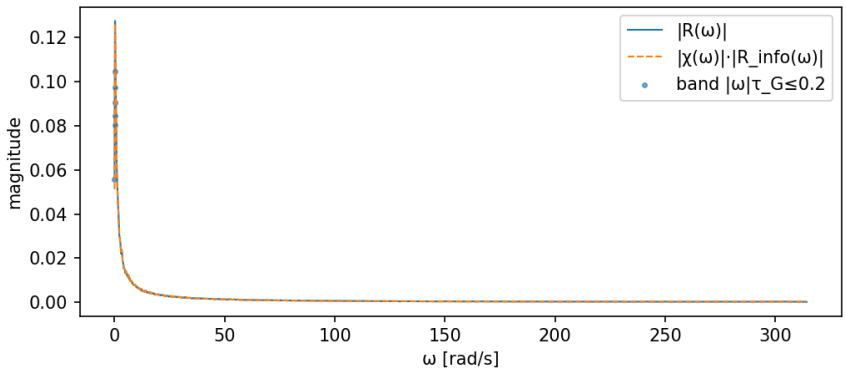


Figure 14: Operational identity: magnitude overlay $|R(\omega)|$ vs $|\chi(\omega)|\,|R_{\text{info}}(\omega)|$; band points $|\omega|\,\tau_G \le 0.2$ marked.

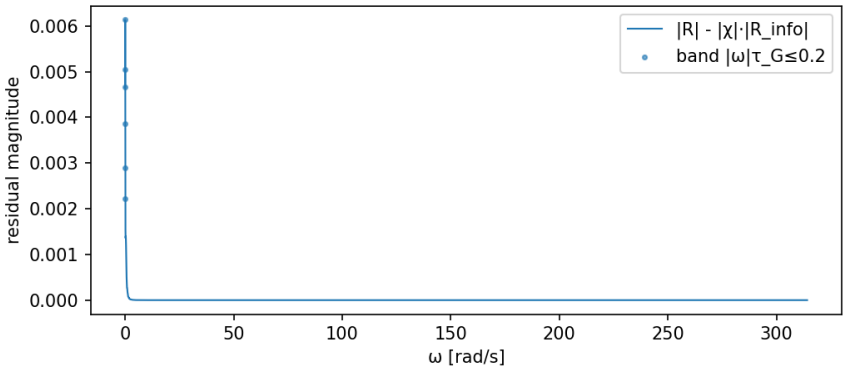


Figure 15: Residual magnitude $||R| - |\chi|\,|R_{\text{info}}||$ vs ω with slow–band markers.

Metrics. The script writes `transfer_identity_metrics.json` with the band size and normalized RMSE, and `transfer_identity_check.csv` with per-frequency values. This directly exhibits the constitutive relation $R = \chi(\Box)R_{\text{info}}$ on the measured slow band. For tensor-channel tests with

projected components (e.g. $A1 \dots A4$) we report also `nrmse_complex_band` and `phase_mae_band` when comparing the empirical complex transfer $S_{R,R_{\text{info}}}/S_{R_{\text{info}},R_{\text{info}}}$ to $\chi(\omega)$ on the same slow band; filenames follow the pattern `transfer_identity_check_tensor_<proj>.csv` and `...metrics_tensor_<proj>.json`.

Appendix C: Assumptions and Tests

Stage / Equation Group	**Core statement or equation**	**Type***	**Assumptions required**	**Empirical / Theoretical test**
Microscopic foundations				
Microscopic process	H1–H3 (local CPTP semigroup; genuine coupling; ergodic non-stationarity)	Necessary	LWSS segment; $C_\Pi \in L^1$; finite variance	Stationarity tests on simulated/experimental $\Pi(t)$
Triad observable	$\Pi_A = \sqrt{F_Q}\, S_A\, I(A{:}\bar{A})$, $[\Pi] = \mathrm{s}^{-1}$	Necessary	Finite F_Q, S, I; well-defined ρ_A	Tomography or entropy/correlation estimators
Positivity + floor	Paley-Zygmund > 0; $\tau_{\text{curv}} := \sqrt{\mathrm{Var}[\Pi]/\langle \dot{\Pi}^2 \rangle}$ diagnostic	Necessary	$\Pi(t) \geq 0$ on measurable set; differentiable	Numerical variance-slope check
OU benchmark	$C_\Pi \in L^1$, $\tau_G = \tau$ (Ornstein–Uhlenbeck)	Theoretical (Lean)	Stationary OU process	`scripts/generate_ou.py`; Appendix B (OU figures, CSV)
GK definition and dispersion (KK)				
Green-Kubo definition	$\tau_G = \int_0^\infty C_\Pi(t)\, dt$	Necessary	Integrable ACF; ergodic	Windowed GK integral ($L \gg \tau_G$)
FDT / Normalization	$\chi(\omega) = \int_0^\infty R(t) e^{-i\omega t} dt$, $R \propto -\frac{d}{dt}\,\mathrm{Cov}$	Necessary	Stationarity; passivity ($\Im\chi \leq 0$)	Verify $\Im\chi$ sign; compare $\chi(0)$ to static susceptibility
GK→KK link	$\chi(\omega) = \chi(0)\big[1 - i\omega\tau_G + O(\omega^2)\big]$	Necessary	Causality (retarded R); analytic χ	Check low-ω slope $= -\tau_G$ from $\chi''(\omega)$;
`scripts/compute_gk_kk.py`				
Debye closure	$\chi(\omega) = (1 + i\omega\tau_G)^{-1}$; $G_{\text{eff}} = G_0\, \chi$	Optional closure	Single-pole kernel; weak long-time tail	Fit spectral data; KK consistency;
`scripts/fit_debye.py`				
Physical scales	$\omega_G = 1/\tau_G$, $\lambda_G = 2\pi c \tau_G$, $m_G = \hbar/(c^2 \tau_G)$	Kinematic	Finite τ_G	Derived from τ_G
Informational geometry				
QFI bound	$\tau_G \geq \hbar/(2\Delta H)$ (via $\sqrt{F_Q} = 2\Delta H/\hbar$)	Optional	Near-unitary regime; fixed H	Compare τ_G to energy uncertainty
Informational curvature	$R^{(\text{info})}_{\mu\nu} = \mathrm{Ricci}[g^{(\text{QFI})}]$	Necessary	Smooth $\rho(\xi)$; finite F_Q	Numerical computation from QFI metric
QFI pull-back example	Induced metric varies with QFI fluctuations (qubit toy)	Demonstrative	Explicit $g^{(\text{QFI})}$; linear embedding	Direct evaluation (Example 1)

Table 7: Summary of assumptions and tests (microscopic foundations, GK definition and dispersion, and informational geometry).

Stage / Equation Group	Core statement or equation	Type*	Assumptions required	Empirical / Theoretical test
Constitutive law and conservation				
Constitutive law	$R_{\mu\nu} = \chi(\Box) R_{\mu\nu}^{(\mathrm{info})}$, $\chi(\omega) = 1 - i\omega\tau_G + \cdots$	Necessary identification	CTP-symmetric kernel; $\chi(0) = 1$	Compare low-ω slope with GK estimate
CTP action	Eq. (C1) $\Rightarrow$ $\nabla^\mu[G_{\mathrm{eff}}(\Box)T_{\mu\nu}] = 0$	Theoretical closure	Retarded analytic R_τ; equal kernels on $\pm$ branches	Formal proof (Sec. 15.8.1)
Conservation law	$\nabla^\mu[G_{\mathrm{eff}}(\Box)T_{\mu\nu}] = 0$	Necessary	CTP symmetry; passivity	Identity (Eq. C4)
Linearized dynamics				
Linearized metric	$\Box\,\bar{h}_{\mu\nu}(\omega,\mathbf{k}) = (16\pi G_0/c^4)\,(1 + i\omega\tau_G)^{-1} T_{\mu\nu}(\omega,\mathbf{k})$	Necessary low-order	Weak-field; harmonic gauge	Verify Debye attenuation/phase lag (Example 2)
Green's function	$\bar{h}(t) = \int_0^\infty ds\,(\tau_G^{-1} e^{-s/\tau_G})\,\bar{h}_{\mathrm{inst}}(t-s)$ (schematic causal memory)	Empirical check	LTI medium; τ_G constant	Compare phase/amplitude to $\chi''(\omega)$
Dispersion law	Debye attenuation: $\lvert G_{\mathrm{eff}}(\omega)\rvert/G_0 = [1 + (\omega\tau_G)^2]^{-1/2}$	Derived prediction	Linear regime; small damping	Phase-shift measurements
Operational pipeline				
Empirical pipeline	$\Pi(t) \to C_\Pi(t) \to \tau_G \to \chi(\omega) \to G_{\mathrm{eff}}(\omega)$	Operational test	Finite-length LWSS window	GK + KK slope consistency $\tau_{\mathrm{KK}} \approx \tau_G$
Transfer identity (slow band) `transfer_identity_mag.png`, `transfer_residual_mag.png`	$R(\omega) \simeq \chi(\omega)\,R_{\mathrm{info}}(\omega)$	Empirical check	$\lvert\omega\rvert\,\tau_G \le 0.2$; LWSS	`scripts/verify_transfer_identity.py`;

Table 8: Summary of assumptions and tests (constitutive law, linearized dynamics, and operational pipeline).

Legend (for Type* column).

- **Necessary**: must hold in linear, causal, passive regimes; failure falsifies the framework at that operating point.
- **Optional closure**: modeling choice (e.g., Debye single pole); may be replaced by higher-order kernels.
- **Empirical check**: measurable procedure used to verify the item.

Appendix D: Derivations and Proofs

Projector conventions and intent

Role. The tensors $\mathcal{P}^{\mu\nu}{}_{\alpha\beta}$ used in the tensor lift are kinematic reshaping operators: they select symmetric (and, when stated, trace-adjusted) components consistent with the slow-band diagnostic. When identity is used instead of a projector, the lowest-order structure of $G_{\mathrm{eff}}(\Box) = G_0\,\chi(\Box)$ is unchanged because the scalar susceptibility multiplies any fixed, local reshaping. In projected channels we report results both componentwise and after recombination; alternative decompositions agree at leading (slow-band) order.

Existence of the Green–Kubo Time τ_G

Theorem 6 (Existence of τ_G under LWSS and mixing)**.** *Let $C : \mathbb{R}_{\ge 0} \to \mathbb{R}$ be the normalized autocorrelation of a band–local, wide–sense stationary (LWSS) observable with bounded sec-*

ond moment. Assume either (i) $C \in L^1(\mathbb{R}_{\geq 0})$ (operational criterion), or (ii) a mixing rate with $|C(t)| \leq c\,(1+t)^{-1-\epsilon}$ for some $\epsilon > 0$. Then the Green–Kubo integral

$$\tau_G \;:=\; \int_0^\infty C(t)\,dt$$

exists and is finite.

Proof sketch. LWSS provides time–translation invariance of second moments; mixing yields decay of the autocovariance. The explicit bound in (ii) gives absolute integrability; (i) is the direct operational criterion reported in Appendix A.

GK→KK Necessity: Small–Frequency Slope

Theorem 7 (GK→KK low–ω slope)**.** *Let R be a retarded response kernel with susceptibility $\chi(\omega) = \int_0^\infty R(t)e^{-i\omega t}dt$. If $t\,R(t) \in L^1(\mathbb{R}_{\geq 0})$ and $C \in L^1(\mathbb{R}_{\geq 0})$ with $\tau_G = \int_0^\infty C(t)dt$, then*

$$\lim_{\omega\to 0} \frac{\Im\chi(\omega)}{\omega} \;=\; -\tau_G.$$

Proof sketch. First fix $T > 0$ and apply dominated convergence on $[0,T]$ to $\int_0^T R(t)\,\frac{\sin(\omega t)}{\omega}\,dt \to \int_0^T t\,R(t)\,dt$. Then let $T \to \infty$, using $tR \in L^1$ to pass to the improper limit. Identify the Green–Kubo time via the fluctuation–dissipation normalization linking R and C.

DC Normalization via Regression/FDT

Lemma 5 (FDT/regression normalization at DC)**.** *Under standard regression/FDT normalization for the retarded kernel R associated with the observable, one has $\chi(0) = \int_0^\infty R(t)\,dt = 1$ and hence $G_{\text{eff}}(0) = G_0$.*

Proof sketch. In the quantum Kubo setting $R(t) \propto \theta(t)\,\langle[A(t), A(0)]\rangle$ and the symmetrized covariance $C(t) \propto \langle\{A(t), A(0)\}\rangle - \langle A\rangle^2$; in classical regression $R(t) \propto -\theta(t)\,\frac{d}{dt}\mathrm{Cov}(A(t), A(0))$. The normalization is fixed so that $\int_0^\infty R(t)\,dt = 1$, hence $\chi(0) = 1$ and $G_{\text{eff}}(0) = G_0$.

Short–time curvature (local bound)

For differentiable $\Pi(t)$ with finite variance and $\langle\dot{\Pi}^2\rangle$, the local Taylor bound

$$\tau_{\text{curv}} \;:=\; \sqrt{\frac{\mathrm{Var}[\Pi]}{\langle\dot{\Pi}^2\rangle}}$$

controls the minimal curvature scale of $C_\Pi(t)$ near $t = 0$ (units: time). In the OU example (Appendix B), this diagnostic scale matches the reported numerical estimate within rounding.

Passivity of Admissible Causal Kernels

Lemma 6 (Passivity)**.** *If R is retarded and of positive type (Bochner), then $\Im\chi(\omega) \leq 0$ for $\omega > 0$. In particular, Debye kernels satisfy passivity, and more generally any admissible kernel in this class does.*

Proof sketch. Standard harmonic analysis: positive–type kernels yield nonpositive imaginary part of the susceptibility on the positive frequency axis.

Curved–Space Commutator Control

Lemma 7 (Commutator bound and $O(\ell_G^2)$ remainder)**.** *For a tensor field $T_{\mu\nu}$ on a smooth background, one has schematically*

$$[\nabla, \Box^n]T \;=\; \mathcal{O}(R\cdot\nabla T),$$

and similarly $[\nabla, (u^\alpha\nabla_\alpha)^n]T = \mathcal{O}(R\cdot\nabla T) + \mathcal{O}((\nabla u)\cdot\nabla T)$. Thus in the slow-band truncation of $(1-\tau_G\, u^\alpha\nabla_\alpha)^{-1}$ the conservation law reads

$$\nabla^\mu[(\mathrm{id} + \tau_G\, u^\alpha\nabla_\alpha + \cdots)\, T_{\mu\nu}] \;=\; \mathcal{O}((\tau_G\nabla)^2),$$

with constants controlled by curvature scales. In flat space, the commutator vanishes and the identity is exact.

Proof sketch. Use standard commutation identities between covariant derivatives and wave operators; collect curvature terms and bound the remainder in the small-ℓ_G (slowly varying) regime.

Bloch-Qubit QFI Pull-Back and Metric Induction

Setup. Consider a single qubit with Hamiltonian $H = \frac{\hbar\omega_0}{2}\,\sigma_z$ and Bloch state

$$\rho(\theta,\phi) = \tfrac{1}{2}\big(I + r\,\mathbf{n}(\theta,\phi)\cdot\boldsymbol{\sigma}\big), \qquad \mathbf{n} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta), \quad r\in(0,1].$$

The QFI metric for this family is [9]

$$\boxed{ds^2 := \frac{dr^2}{1-r^2} + r^2\,(d\theta^2 + \sin^2\theta\, d\phi^2)} \tag{90}$$

For constant purity r, only the angular part contributes; for the generator H one has $F_Q(\rho;H) = 4(\Delta H)^2/\hbar^2 = r^2\omega_0^2$.

Pull-back. Embed by a linear map $\xi^i(x^\mu) = (\theta(x), \phi(x))$ with constant $\partial_\mu\xi^i = k^i_{;\mu}$. The pulled-back metric is

$$\boxed{g_{\mu\nu}^{(\mathrm{geo})} := \partial_\mu\xi^i\,\partial_\nu\xi^j\, g_{ij}^{(\mathrm{QFI})} := \tfrac{1}{4}F_Q(\rho;H)\,k_\mu k_\nu := \frac{r^2\omega_0^2}{4}\,k_\mu k_\nu} \tag{91}$$

Variation and triad link. A small fluctuation $r\to r+\delta r$ changes the QFI as $\delta F_Q = 8r\,\delta r\,\omega_0^2/\hbar^2$. Identifying the triad amplitude

$$\Pi := \sqrt{F_Q}\,S\,I = \frac{2\omega_0 r}{\hbar}\,S\,I, \tag{92}$$

gives, to leading order,

$$\delta g_{\mu\nu}^{(\mathrm{geo})} = \tfrac{1}{4}\,\delta F_Q\,k_\mu k_\nu, \tag{93}$$

illustrating how informational fluctuations modify the induced metric in this toy setting; macroscopic response is through the causal kernel $\chi(\omega)$.

Scales. For relaxation rate γ: $\tau_G \simeq \gamma^{-1}$, $\omega_G = \gamma$, $\lambda_G = 2\pi c/\gamma$, $m_G = \hbar\gamma/c^2$, realizing $\chi(\omega) = (1+i\omega\tau_G)^{-1}$ and $G_{\mathrm{eff}}(\omega) = G_0\,\chi(\omega)$.

Linearized Metric Response and Retarded Green's Function

Screened operator. From $R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = (8\pi/c^4)\, G_{\text{eff}}(\Box)\, T_{\mu\nu}$, in harmonic gauge the components obey

$$\boxed{\Box\, \bar{h}_{\mu\nu}(\omega, \mathbf{k}) \;=\; \frac{16\pi G_0}{c^4} \frac{1}{1 + i\omega\tau_G} T_{\mu\nu}(\omega, \mathbf{k}), \quad \ell_G := c\,\tau_G.} \tag{D2.2}$$

in the local rest frame (slow-band representative), exhibiting Debye attenuation and phase lag governed by τ_G.

Retarded kernel. In time domain, the same slow-band closure corresponds to convolving the source with a retarded exponential kernel in time, producing causal memory with decay rate τ_G^{-1} (cf. Sec. 15.8.1).

Dispersion and conservation. The closure above is relaxational (Maxwell–Cattaneo/Debye) rather than a Proca mass term: it attenuates high-frequency response through $1/(1 + i\omega\tau_G)$ while leaving the low-frequency propagation structure intact. An action-level construction (Sec. 15.8.1) yields $\nabla^\mu[G_{\text{eff}}(\Box)(T^{(\text{triad})} + T^{(\text{grav})})_{\mu\nu}] = 0$.
Interpretation. The persistence scale τ_G (equivalently $\ell_G = c\,\tau_G$) sets the causal latency of the response; it is not a graviton mass.

Appendix E: Lean proof status (summary)

Paper claim	Lean file / theorem	Status
GK→KK small-ω slope	`KK.lean`: `dct_finite_window_integral`	Proven (GK ⇒ KK slope)
Assembly theorem (finite window)	`NecessityTheorem.lean`: `kkSlopeTrunc_tendsto_zero_to_moment`	Proven (paper-style wrapper)
Improper GK→KK slope ([0,∞))	`NecessityImproper.lean`: `kkSlopeImproper_tendsto_zero_to_moment`	Proven (half-line upgrade under $tR \in L^1$)
Debye sufficiency $\chi = (1 + i\omega\tau)^{-1}$	`Debye.lean`: `truncTransform_tendsto_chiDebye`, `chi_passivity_nonpos`	Proven (sufficiency + passivity)
Flat-space conservation $\nabla[G(\Box)T] = 0$	`Flat.lean`: `conservation_poly`	Proven (polynomial $G(\Box)$ preserves conservation)
Convolution/CTP conservation	`Convolution.lean`: `convolution_preserves_conservation_pointwise`	Proven (translation-invariant kernel)
Wiener–Khinchin $S(0) = 2\tau_G$	`WK.lean`: `psd0_trunc_tendsto_of_gk_trunc`	Proven (DC PSD limit)
OU: $C(t) \in L^1$, $\tau_G = \tau$	`OU.lean`: `ou_acf_in_L1`, `ou_tau_G_eq_tau`	Proven (OU ACF integrable; $\tau_G = \tau$)
Paley–Zygmund positivity	`Support.lean`: `paley_zygmund_simple`	Proven (existence lemma)
$\tau_G > 0$ (measure theory)	`Support.lean`: `tau_G_positive`	Proven (positivity of GK integral)
L^1 under H1–H3 (general)	`Support.lean`: `TriadHypotheses`	Assumed / documented
Short-time diagnostic scale τ_{curv}	–	Not formalized here (paper-level diagnostic)

References

[1] Ovidiu Tataru. Geometric response and the frequency–dependent gravitational coupling. *International Journal of Quantum Foundations*, 11(4):853–905, 2025. Also available at https://ijqf.org/archives/7742.

[2] Robert M. Wald. *General Relativity*. University of Chicago Press, 1984.

[3] Sean M. Carroll. *Spacetime and Geometry: An Introduction to General Relativity*. Addison-Wesley, 2004.

[4] Leo P. Kadanoff and Paul C. Martin. Hydrodynamic equations and correlation functions. *Annals of Physics*, 24(3):419–469, 1963.

[5] Ovidiu Tataru. A quantum structural triad: Fluctuations, entropy, and correlations as interdependent primitives. *International Journal of Quantum Foundations*, 11(4):800–852, 2025. Also available at https://ijqf.org/archives/7737.

[6] T. Jacobson. Thermodynamics of spacetime: The einstein equation of state. *Physical Review Letters*, 75:1260–1263, 1995.

[7] Ryogo Kubo. Statistical-mechanical theory of irreversible processes. i. general theory and simple applications to magnetic and conduction problems. *Journal of the Physical Society of Japan*, 12(6):570–586, 1957.

[8] Robert Zwanzig. *Nonequilibrium Statistical Mechanics*. Oxford University Press, 2001.

[9] Samuel L. Braunstein and Carlton M. Caves. Statistical distance and the geometry of quantum states. *Physical Review Letters*, 72(22):3439–3443, 1994.

[10] Sebastian Deffner and Eric Lutz. Energy–time uncertainty relation for driven quantum systems. *Journal of Physics A: Mathematical and Theoretical*, 46:335302, 2013.

[11] C. W. Helstrom. *Quantum Detection and Estimation Theory*. Academic Press, 1976.

[12] D. Bures. An extension of kakutani's theorem on infinite product measures to the tensor product of semifinite w^*-algebras. *Transactions of the American Mathematical Society*, 135:199–212, 1969.

[13] Dénes Petz. Monotone metrics on matrix spaces. *Linear Algebra and its Applications*, 244:81–96, 1996.

[14] Shun ichi Amari. *Information Geometry and Its Applications*. Springer, 2016.

[15] Esteban A. Calzetta and Bei-Lok Hu. *Nonequilibrium Quantum Field Theory*. Cambridge University Press, 2008.

[16] Julian Schwinger. Brownian motion of a quantum oscillator. *Journal of Mathematical Physics*, 2:407–432, 1961.

[17] A. Kamenev. *Field Theory of Non-Equilibrium Systems*. Cambridge University Press, 2011.

[18] P. D. Welch. The use of fast fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms. *IEEE Transactions on Audio and Electroacoustics*, 15(2):70–73, 1967.

[19] T. Damour. Black-hole eddy currents. *Physical Review D*, 18:3598–3604, 1978.

[20] K. S. Thorne, R. H. Price, and D. A. Macdonald. *Black Holes: The Membrane Paradigm*. Yale University Press, 1986.

[21] J. D. Bekenstein. Black holes and entropy. *Physical Review D*, 7:2333–2346, 1973.

[22] S. W. Hawking. Particle creation by black holes. *Communications in Mathematical Physics*, 43:199–220, 1975.

[23] P. Kovtun, D. T. Son, and A. O. Starinets. Viscosity in strongly interacting quantum field theories from black-hole physics. *Physical Review Letters*, 94:111601, 2005.

[24] T. Faulkner, A. Lewkowycz, and J. Maldacena. Quantum corrections to holographic entanglement entropy. *Journal of High Energy Physics*, 11:074, 2013.

[25] E. Berti, V. Cardoso, and A. O. Starinets. Quasinormal modes of black holes and black branes. *Classical and Quantum Gravity*, 26:163001, 2009.

[26] T. Jacobson and D. Mattingly. Gravity with a dynamical preferred frame. *Physical Review D*, 64:024028, 2001.

[27] T. Padmanabhan. Thermodynamical aspects of gravity: New insights. *Reports on Progress in Physics*, 73:046901, 2010.

[28] A. D. Sakharov. Vacuum quantum fluctuations in curved space and the theory of gravitation. *Soviet Physics Doklady*, 12:1040, 1968.

[29] S. Deser and R. P. Woodard. Nonlocal cosmology. *Physical Review Letters*, 99:111301, 2007.

[30] M. Maggiore and M. Mancarella. Nonlocal gravity and dark energy. *Physical Review D*, 90:023005, 2014.

[31] Y. Dirian, S. Foffa, N. Khosravi, M. Kunz, and M. Maggiore. Nonlocal gravity and dark energy phenomenology. *Journal of Cosmology and Astroparticle Physics*, 06:033, 2014.

[32] Y. Dirian, S. Foffa, M. Kunz, M. Maggiore, and V. Pettorino. Non-local gravity and comparison with observational datasets. *Journal of Cosmology and Astroparticle Physics*, 05:068, 2016.

[33] R. P. Woodard. Nonlocal models of cosmic acceleration. *Foundations of Physics*, 44:213–233, 2014.

[34] Y.-F. Cai, S. Capozziello, M. De Laurentis, and E. N. Saridakis. f(t) teleparallel gravity and cosmology. *Reports on Progress in Physics*, 79:106901, 2016.

[35] F. Dowker. Causal sets and the deep structure of spacetime. *Contemporary Physics*, 47:1–9, 2006.

[36] J. F. Donoghue and B. K. El-Menoufi. Nonlocal quantum effects in cosmology: Quantum memory, nonlocal actions, and gravity. *Journal of High Energy Physics*, 10:044, 2015.

[37] B. P. Abbott et al. Gw170817: Observation of gravitational waves from a binary neutron star inspiral. *Physical Review Letters*, 119:161101, 2017.

[38] M. Crossley, P. Glorioso, and H. Liu. Effective field theory of dissipative fluids. *Journal of High Energy Physics*, 09:095, 2017.

[39] R. Abbott et al. Gwtc-3: Compact binary coalescences observed by LIGO and Virgo during the second part of the third observing run. *Physical Review X*, 13:041039, 2023.

[40] R. Abbott et al. Tests of general relativity with gwtc-3. *Physical Review D*, 112:084080, 2025.

[41] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration. Open data from LIGO, Virgo, and KAGRA through the first part of the fourth observing run. *Astrophysical Journal Letters*, 2025. in press.

[42] S. Mastrogiovanni, L. Haegel, C. Karathanasis, I. Magaña Hernandez, and D. A. Steer. Gravitational wave friction in light of GW170817 and GW190521. *Journal of Cosmology and Astroparticle Physics*, 02:043, 2021.

[43] J. M. Weisberg and Y. Huang. Relativistic measurements from timing the binary pulsar psr b1913+16. *The Astrophysical Journal*, 829:55, 2016.

[44] NANOGrav Collaboration. The nanograv 15-year data set. Zenodo, 2023.

[45] R. E. A. C. Paley and A. Zygmund. A note on analytic functions in the unit circle. *Proceedings of the Cambridge Philosophical Society*, 28:266–272, 1932.

[46] G. D. Birkhoff. Proof of the ergodic theorem. *Proceedings of the National Academy of Sciences of the USA*, 17:656–660, 1931.

[47] R. S. Hamilton. Three-manifolds with positive ricci curvature. *Journal of Differential Geometry*, 17:255–306, 1982.

[48] D. M. DeTurck. Deforming metrics in the direction of their ricci tensors. *Journal of Differential Geometry*, 18:157–162, 1983.

[49] P. J. Mohr, D. B. Newell, B. N. Taylor, and E. Tiesinga. Codata recommended values of the fundamental physical constants: 2022. *Reviews of Modern Physics*, 97:025002, 2025.

[50] H. M. Nussenzveig. *Causality and Dispersion Relations*. Academic Press, 1972.

[51] E. C. Titchmarsh. *Introduction to the Theory of Fourier Integrals*. Oxford University Press, 1948.

[52] H. B. Callen and T. A. Welton. Irreversibility and generalized noise. *Physical Review*, 83(1):34–40, 1951.

[53] Norbert Wiener. Generalized harmonic analysis. *Acta Mathematica*, 55:117–258, 1930.

[54] A. Khintchine. Korrelationstheorie der stationären stochastischen prozesse. *Mathematische Annalen*, 109:604–615, 1934.

[55] W. Pusz and S. L. Woronowicz. Passive states and kms states for general quantum systems. *Communications in Mathematical Physics*, 58:273–290, 1978.

[56] L. V. Keldysh. Diagram technique for nonequilibrium processes. *Soviet Physics JETP*, 20:1018–1026, 1965.

[57] P. Debye. *Polar Molecules*. The Chemical Catalog Company, 1929.

[58] C. Cattaneo. Sulla conduzione del calore. *Atti Sem. Mat. Fis. Univ. Modena*, 3:83–101, 1948.

[59] P. Vernotte. Les paradoxes de la théorie continue de l'équation de la chaleur. *Comptes Rendus*, 246:3154–3155, 1958.

[60] H. Mori. Transport, collective motion, and brownian motion. *Progress of Theoretical Physics*, 33(3):423–455, 1965.

[61] R. Zwanzig. Memory effects in irreversible thermodynamics. *Physical Review*, 124(4):983–992, 1961.

[62] G. E. Uhlenbeck and L. S. Ornstein. On the theory of the brownian motion. *Physical Review*, 36:823–841, 1930.

[63] W. Israel and J. M. Stewart. Transient relativistic thermodynamics and kinetic theory. *Annals of Physics*, 118(2):341–372, 1979.

[64] W. A. Hiscock and L. Lindblom. Generic instabilities in first-order dissipative relativistic fluid theories. *Physical Review D*, 31(4):725–733, 1985.

[65] Rick Durrett. *Probability: Theory and Examples*. Cambridge University Press, 5 edition, 2019.

[66] Bruno Nachtergaele and Robert Sims. Lieb–robinson bounds in quantum many-body physics. In *Entropy and the Quantum*, volume 529 of *Contemporary Mathematics*, pages 141–176. American Mathematical Society, 2011.